

UNIVERSAL MULTIMODAL CO-GENERATION WITH CONTINUOUS BITSTREAM DIFFUSION

Anonymous authors

Paper under double-blind review

ABSTRACT

[Luca will lead the pitch and framing of the paper; to be written last, once results are final.]

1 INTRODUCTION

• **Micha:** See my suggested intro below

Continuous-state generative diffusion models (Ho et al., 2020; Song et al., 2021b) have achieved strong results in image (Rombach et al., 2022b), video (Ho et al., 2022), and audio generation (Kong et al., 2021). These models generate samples by learning to reverse a corruption process that progressively transforms data into noise. Their success has motivated extending diffusion to discrete domains, including language (Sahoo et al., 2024) and protein sequences (Wang et al., 2024b).

For discrete data, widely studied approaches use discrete corruption processes, including masking (Sahoo et al., 2024) and categorical token replacement (Austin et al., 2021). An alternative is to represent discrete tokens in a continuous space and apply Gaussian diffusion, as explored by Diffusion-LM (Li et al., 2022) and continuous diffusion for categorical data (Dieleman et al., 2022). Analog Bits provides a particularly simple representation: encode discrete values as binary strings and treat their bits as continuous variables during diffusion (Chen et al., 2023a). Recent continuous diffusion and flow-based language models report sample quality competitive with, and in some settings better than, discrete diffusion baselines (Chen et al., 2026; Hu et al., 2026). These results motivate investigating continuous diffusion as a shared modeling framework for heterogeneous modalities.

Multimodal generation requires capturing dependencies between observations with different statistical and structural properties, such as images and captions, speech and transcripts, or protein sequences and structures. Existing diffusion and flow-based systems already support joint and conditional generation in image–text settings (Bao et al., 2023b; Tang et al., 2023b) and protein sequence–structure settings (Campbell et al., 2024; Wang et al., 2025b). Building on this progress, we investigate how far a common representation, denoising objective, and data-dependent calibration rule can transfer across these domains, with modality-specific choices concentrated in tokenization and data layout.

In this paper, we extend continuous bitstream diffusion (CoBit) from language modeling to multimodal co-generation (Batzolis et al., 2026a). Frozen modality-specific tokenizers map observations to discrete codes, which are organized into a shared bitstream and corrupted with continuous Gaussian noise. For each modality combination, a single denoiser is trained on joint and conditional tasks through partial denoising: observed positions remain clean while the remaining positions are corrupted and reconstructed. The same weights therefore support joint generation and the conditional directions included in training. The extension introduces a shared multimodal code space and a partial-denoising training recipe while retaining CoBit’s corruption process, denoising objective, and backbone family.

Different modality combinations can lose information at different noise levels, motivating data-dependent calibration. Following text-only CoBit, we estimate a conditional-entropy-rate profile from denoising errors during training and use it to guide training noise allocation (Raya et al., 2026) and the placement of inference steps (Stancevic et al., 2025). The resulting schedule provides a common calibration rule across domains. Tokenizer configurations, sequence layouts, and inference-time settings such as guidance, stochasticity, and step count remain task-dependent.

We evaluate this framework on image–text, speech–text, and protein sequence–structure tasks. On zero-shot MS-COCO evaluation, the 462M-parameter image–text model achieves a text-to-image FID-30K of 18.48 with 20 denoiser evaluations, while also supporting captioning and joint generation from the same weights. The 633M-parameter speech–text model achieves a 3.7% word error rate between co-generated text and Whisper transcriptions of the generated speech. Protein experiments assess folding, inverse folding, and generated-backbone designability. Together, these experiments examine the transfer of a shared bitstream diffusion recipe across domains and identify the task-dependent performance and sampling trade-offs that remain.

2 RELATED WORK

[To be written.]

3 METHOD

Our model is the continuous bitstream diffusion model of Batzolis et al. (2026b) (CoBit), a text-generation framework, applied *unchanged* to sequences that interleave several modalities. The corruption process, the denoiser parameterisation and its loss, the noise schedule and its estimator, the self-conditioning mechanism, the sampler and the backbone are those of the text-only model, with the same hyperparameters. What is added is small and lives entirely in the data pipeline: (i) a way of writing any set of tokenised modalities into one shared binary vocabulary, so that a single network reads and writes all of them through one interface (§3.2); and (ii) a training recipe under which one set of weights supports unconditional co-generation of all modalities present, conditional generation in every direction between them, and any mixture of the two, with the direction chosen at inference time by deciding which positions are held fixed (§3.3). There are no modality-specific encoders, decoders, heads, embedding tables or losses, and no modality-specific training stages. We recap the base model first, to the extent needed to make these two additions precise.

3.1 CONTINUOUS BITSTREAM DIFFUSION

Bits as the universal representation. CoBit models a sequence of N discrete tokens $t_1, \dots, t_N$ from a vocabulary of size V by writing each token as its $m = \lceil \log_2 V \rceil$ -bit binary index, giving a flat bitstream $\mathbf{x}_0 \in \{0, 1\}^S$ with $S = Nm$. Nothing else about the data is used: the framework does not know whether a token is a word piece, an image patch code, a speech frame or a residue, and it never sees a token identity, only its bits. Once written as bits, the sequence is treated as continuous data. A standard variance-exploding Gaussian diffusion is run on it, $\mathbf{x}_\sigma = \mathbf{x}_0 + \sigma \epsilon$ with $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\sigma \in [\sigma_{\min}, \sigma_{\max}] = [0.002, 80]$, taking the data centre to be $1/2$ and $\sigma_{\text{data}} = 1/2$. A denoiser $D_\theta(\mathbf{x}_\sigma, \sigma) \in (0, 1)^S$ predicts the posterior probability that each bit is one and is trained with the EDM-weighted denoising loss (Karras et al., 2022) in probability space,

$$\mathcal{L}(\theta) = \mathbb{E} \left[w(\sigma) \left\| D_\theta(\mathbf{x}_\sigma, \sigma) - \mathbf{x}_0 \right\|^2 \right], \quad w(\sigma) = \frac{\sigma^2 + \sigma_{\text{data}}^2}{\sigma^2 \sigma_{\text{data}}^2}, \quad (1)$$

which induces the score $\nabla \log p_\sigma(\mathbf{x}_\sigma) \approx (D_\theta - \mathbf{x}_\sigma)/\sigma^2$ used by the sampler. The network is not asked to learn the local Gaussian denoising problem: the analytic posterior logit of an isolated bit under a uniform prior, $(\mathbf{x}_\sigma - \frac{1}{2})/\sigma^2$ (clipped to ± 30), is added to a learned contextual residual, so that the network models only the dependence *between* bits—which is where all modality-specific structure resides. Three components of the base recipe matter for what follows.

Entropy-derived noise schedule. The rate at which information about $\mathbf{x}_0$ is destroyed as σ grows depends on the data, and different modalities—and mixtures of modalities—lose their information at different noise levels. Rather than fixing a proposal over σ by hand, CoBit measures this rate on the data being modelled and allocates both training noise levels and sampling steps according to it. The conditional entropy rate per unit $\log \sigma$ is estimated online from the unweighted per-example denoising error, $\hat{h}(\sigma) \propto e(\sigma)/\sigma^2$ with $e(\sigma) = \frac{1}{S} \|D_\theta - \mathbf{x}_0\|^2$, kept in a FIFO buffer over recent training examples (8×10^5) and binned into 128 log-spaced bins. The training proposal for σ , and through its inverse CDF the sampling grid, is $q(\sigma) \propto g(\sigma; c, n) \hat{h}(\sigma)^{1/2}$, where $g(\sigma; c, n) = \sigma^n / (\sigma^n + c^n)$ ($c = 0.1$, $n = 3$) is a smooth gate removing mass from very small σ . Training

starts from the EDM log-normal proposal ($P_{\text{mean}} = -1.2$, $P_{\text{std}} = 1.2$), ramps linearly to q over steps $4\text{--}5 \times 10^4$, and refreshes q every 2×10^3 steps thereafter. Because $\hat{h}$ is measured rather than assumed, the schedule is self-calibrating: the same estimator with the same hyperparameters is used for every modality combination, and it adapts itself to each.

Self-conditioning. Following Chen et al. (2023b), the denoiser takes a second input $\hat{x}_0$, an estimate of the clean bits, concatenated bitwise with x_σ . During training, with probability $p_{\text{sc}} = 0.5$ a first no-gradient pass produces $\hat{x}_0 = D_\theta(x_\sigma, \sigma, \mathbf{0})$, which is detached and fed to the second, trained pass; otherwise $\hat{x}_0 = \mathbf{0}$. At sampling time two refresh modes are available. In `carry` mode the probabilities returned at step i are passed as $\hat{x}_0$ to step $i+1$, so self-conditioning costs no additional network evaluations; in `refined` mode an extra denoiser pass is run at the new state $(x_{\sigma_{i+1}}, \sigma_{i+1})$ conditioned on the carried estimate, doubling the evaluations per step. We use `carry` throughout unless stated otherwise.

Sampler and backbone. We integrate the probability-flow ODE $dx/d\sigma = (x - D_\theta(x, \sigma))/\sigma$ with a first-order (DDIM/Euler) step (Song et al., 2021a) on the entropy-derived grid, optionally with EDM-style churn (re-noising by a factor $1 + \gamma_i$ before each step; Karras et al., 2022) as the stochastic variant. Integration stops at a terminal $\sigma_{\text{dec}} > \sigma_{\text{min}}$; the final probabilities are thresholded at $1/2$, regrouped into m -bit codes and handed to the appropriate tokenizer’s decoder. The number of steps, the guidance scale (§3.3) and the churn are inference-time choices that require no retraining. The denoiser is the Sequence Diffusion Transformer of Batzolis et al. (2026b): the m bits of a position, with their per-bit positional features, form one patch that is linearly projected to one trunk token, so the transformer runs at the token length N rather than the bit length Nm ; the trunk uses RoPE, SwiGLU and adaLN-zero conditioning on σ , and a light skip-connection head combines the trunk output with the raw noisy bits to produce the m per-bit logits of each position. The same backbone is used for every modality combination; only its depth and width vary between checkpoints.

3.2 MULTIMODAL BITSTREAMS

The base model needs nothing from its input except that it be a sequence of fixed-width binary codes. This is what makes it modality-agnostic, and it makes extending it to new modalities a data-preparation step rather than a modelling one. Three ingredients are required.

Modality-specific tokenizers. Each modality is mapped to a run of integer codes by a discrete tokenizer of its own—a byte-pair vocabulary for text, a vector- or lookup-free quantiser for images or audio, a residue alphabet or a structure quantiser for proteins—and back by the matching decoder. The tokenizers are the *only* modality-specific components in the system. They are frozen, they are not trained or fine-tuned jointly with the diffusion model, and they sit outside it: the diffusion model never sees an embedding, a spectrogram or a pixel, only bits. Any tokenizer that yields fixed-width codes can be used; quantisers whose codebook is the full set of sign patterns of a latent vector (lookup-free or finite-scalar quantisation; Yu et al., 2024; Mentzer et al., 2024) pair especially naturally with bitstream diffusion, since every bit string is then a valid code and the bits are the quantiser’s own latent coordinates rather than an arbitrary index. The tokenizers used for each modality pair are stated in the corresponding experimental subsection (§4).

One shared code space. The per-modality vocabularies are placed in disjoint ranges of a single integer code space, together with a small reserved block of control codes (padding and segment markers), and m is chosen as the bit width of that union—in practice by offsetting the vocabularies or, equivalently, by prepending one or more namespace bits that select the modality. The m bits of a position therefore determine both the modality and the content of a token, every position is a “token of any modality”, and the patch size of the backbone is m for all of them. Consequently there are no modality-specific embedding tables, encoders, heads or losses, and the shared Bernoulli output head costs $O(m) = O(\log V)$ per position regardless of how many vocabularies are merged. The one modality-aware parameter we add is a learned segment embedding: a zero-initialised vector per segment type (control, text, image, audio, ...) added to the per-bit trunk input, which tells the network which segment a bit belongs to without introducing any per-modality computation.

A fixed sequence template. A training example is laid out as a fixed-length template of segments, one per modality, each of a fixed token budget (truncated or padded), separated by structural markers (start/end of sequence and of each segment). Markers are ordinary codes in the reserved block; they are always presented clean, at training and at sampling, and are never denoised. Two-dimensional or

higher-dimensional token grids are flattened along a locality-preserving (Hilbert) curve so that 1-D adjacency reflects spatial adjacency. Modalities that are aligned position by position (for instance a residue identity and a structure token describing the same site) may also share one position by concatenating their codes into a single m -bit word; this too is only a layout choice. *[Confirm against the final protein codec in §4.5: the current stub describes an 18-bit position formed by a 5-bit residue code and a 13-bit DPLM-2 structure code.]* The transformer sees only the flat bitstream; the same code path decodes each segment through its own tokenizer’s decoder.

What is, and is not, modality-specific. Across every experiment in this paper the corruption process, the σ range and σ_{data} , the loss and its weighting, the entropy-schedule estimator and all of its hyperparameters, self-conditioning, the sampler family and the backbone architecture are identical. *[Verify against the final audio and protein training configurations before submission; this sentence is the paper’s central claim and should be checked field by field.]* What changes between modality pairs is confined to the data pipeline: the tokenizers and their decoders, the segment layout (which modalities, in which order, with what token budgets), and the list of training regimes of §3.3. The only quantities chosen *per direction* at inference are the sampler settings (step count, guidance scale, churn), which are free hyperparameters of any diffusion sampler and involve no retraining and no architectural difference.

3.3 GENERATIVE DIRECTIONS

A sequence built as above is partitioned into segments $S_1, \dots, S_K$ (one per modality, plus markers). Any generation task over these modalities is a choice of a *conditioning set* $\mathcal{C}$ of segments that are given and a *target set* $\mathcal{T}$ that must be produced: $\mathcal{C} = \emptyset$ is unconditional joint co-generation of everything; conditioning on a text segment gives text-to-image or text-to-speech; conditioning on an image or a speech segment gives captioning or transcription; conditioning on a prefix of a segment gives continuation; and so on. CoBit implements every such task with the same network and the same weights through a single mechanism, which we call *partial denoising*: the positions in $\mathcal{C}$ are held at their clean values x_0 throughout, and diffusion acts only on the positions in $\mathcal{T}$. Because the model is a bidirectional denoiser over the whole sequence, a clean segment is simply part of the input it conditions on; no task token, task-specific head or encoder is involved.

Training. For each training example we draw a task from a small list of *regimes*—conditioning sets with sampling probabilities—and construct the noisy input accordingly. Let $M \in \{0, 1\}^S$ be the bit mask of the conditioned positions (the segments in $\mathcal{C}$, always including the markers). The input to the denoiser is $x_\sigma = M \odot x_0 + (1 - M) \odot (x_0 + \sigma\epsilon)$, i.e. the conditioning segments are clamped clean rather than noised, and the loss equation 1 is averaged only over the denoised positions $1 - M$. When self-conditioning is active the clean conditioning bits are also written into $\hat{x}_0$ on the masked positions, so the network never has to predict what it is given. To enable classifier-free guidance (Ho & Salimans, 2021), with probability $p_{\text{uncond}} = 0.1$ the conditioning segments (but never the markers) are replaced by the null value $1/2$ —the data centre and the point of zero information for a bit under Gaussian corruption—so that the network also learns the unconditional denoiser for each target set. The regime list is the whole specification of which directions a model supports: adding a direction for a given modality set means adding an entry to this list, without touching the model. Each experimental subsection states the regimes used. Examples whose conditioning was dropped are excluded from the entropy-rate buffer of §3.1, so the schedule reflects the conditional denoising problems the model is actually asked to solve.

Sampling. At inference the task fixes M and the clean values on the conditioned positions; the sampler of §3.1 initialises the target positions from noise, clamps the conditioned positions to their clean bits before every denoiser call, zeroes the ODE velocity on those positions after every call (so they never move) and clamps them once more in the returned probabilities. Guidance uses the standard two-branch estimate in probability space: a conditional branch sees the clean conditioning, an unconditional branch sees the null value $1/2$ on the same positions (markers protected), the two are evaluated as one batch of size $2B$, and the guided estimate $D_g = D_u + w(D_c - D_u)$ replaces D_θ in the score, where $w = 1$ recovers the plain conditional model and $w > 1$ sharpens it. In *carry* mode each branch carries its own self-conditioning estimate. Joint co-generation ($\mathcal{C} = \emptyset$) uses a single branch, so its cost per step is one network evaluation while every guided conditional direction costs two; we report the number of function evaluations (NFE) accordingly. Since the direction is decided entirely by M , the remaining sampler choices—step count, guidance scale,

Modality	Tokenizer	Quantization Scheme	Vocabulary Size
<i>Class-conditional ImageNet</i>			
<i>Image and Text</i>			
Image	OpenMAG-VIT2 (Luo et al., 2024)	LFQ	—
Text	$\circ 200k$	Byte-level BPE	200,018
<i>Audio and Text</i>			
Audio	StableCodec (Parker et al., 2025)	FSQ	46,656
Speaker	BiCodec (Wang et al., 2025a)	FSQ	4,096
Text	$\circ 200k$	Byte-level BPE	200,018
<i>Protein Sequence and Structure</i>			
Amino-acid	—	—	32
—	DPLM-2 (Wang et al., 2024a)	LFQ	—

Table 1: Overview of tokenization methods used across the different experiments.

churn—can be set per direction on the same weights, and nothing in the framework requires them to agree across directions. The same weights therefore act as an unconditional co-generator, as a family of conditional generators in every direction seen in training, and as a model that fills in an arbitrary subset of positions given the rest.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

[Shared setup common to all modalities.]

4.2 IMAGE GENERATION (CLASS-CONDITIONAL IMAGENET)

[Georgios to write: image-only results and comparisons.]

4.3 IMAGE AND TEXT (CC3M, CC12M)

Setup. We train CoBit-CC3M (462M parameters, 24×1024) on Conceptual Captions 3M (Sharma et al., 2018) ($\approx 2.9M$ pairs, 10^6 steps) and CoBit-CC12M (633M, 26×1152) on Conceptual 12M (Changpinyo et al., 2021) ($\approx 11.7M$ pairs, 1.744×10^6 steps, 87% of its planned budget). Images are tokenized by a frozen Open-MAGVIT2 lookup-free quantiser (Luo et al., 2024; Yu et al., 2024) into 18-bit codes, $16 \times 16 = 256$ tokens ($f16$) for CC3M and $32 \times 32 = 1,024$ ($f8$) for CC12M; text by the $\circ 200k$ byte-pair vocabulary (caption budgets of 48 and 96 tokens). A 19th namespace bit separates image from text and control codes, so every position is one 19-bit token (§3.2); training regimes are 0.35 text-conditioned, 0.35 image-conditioned, 0.30 joint. Neither model sees MS-COCO (Lin et al., 2014): all numbers are zero-shot on COCO val2014 under one fixed protocol (FID-30K, CMMD (Jayasumana et al., 2024), CLIP ViT-B/32 (Radford et al., 2021); captioning on the Karpathy test split; Appendix A.1). Deliberately absent are text-only pre-training, tokenizer surgery, auxiliary alignment losses and staged curricula. All are known to help and we expect them to help here; we omit them so that the result measures what the unmodified framework transfers, and leave them to future work.

Results. Table 2 reports all four directions for both models, each from a single set of weights. CC3M reaches zero-shot FID-30K **18.48** at NFE 20. Measured against a disjoint set of real images, as that FID itself is, the floor of the column is 2.94, of which 2.35 is the finite-sample floor any model pays at this n and 0.59 the codec, so the gap to larger systems lies in the model, not in the token space. Joint co-generation is much harder than either conditional direction (FID 50.19 at 10 steps), as with no prompt the model samples its own CC3M-like training distribution, yet the co-generated image and caption agree with each other (pair coherence 25.67 at 256 steps) about as well as a conditional caption agrees with its image (24.83).

Direction	n	Steps (NFE)	FID↓	CMMD↓	CLIP↑	CIDER↑
<i>CoBit-CC3M — one set of weights (462M, f16, 256 image tokens)</i>						
text → image	30,000	10 (20)	18.48	18.41	27.73	—
image → text (CLIP)	5,000	256 (512)	—	—	24.83	—
image → text (captioning)	5,000	256 (512)	—	—	—	0.136
joint, 10 steps	30,000	10 (10)	50.19	41.53	23.10 [†]	—
joint, 256 steps	30,000	256 (256)	61.95	46.80	25.67 [†]	—
<i>CoBit-CC12M — one set of weights (633M, f8, 1,024 image tokens)</i>						
text → image	30,000	16 (32)	16.59	17.76	26.51	—
image → text (CLIP)	5,000	256 (512)	—	—	22.98	—
image → text (captioning)	5,000	256 (512)	—	—	—	0.057
joint, 20 steps	5,000 [‡]	20 (20)	56.18	37.60	23.19 [†]	—
joint, 256 steps	5,000 [‡]	256 (256)	64.47	40.33	24.01 [†]	—
<i>FID floor: two disjoint real draws</i>	30,000		2.35	—	—	—
<i>after the codec, f16 / f8</i>	30,000		2.94 / 2.55	—	—	—

Table 2: Zero-shot MS-COCO, all four generative directions from a single set of weights per model. Each conditional row is quoted at the step count that is best for that direction (*Sampler* below); joint (unconditional) generation has no single optimum, so both ends of its trade-off are shown. CLIP is ViT-B/32, reported as $100 \cdot \cos$; per-direction guidance and churn are listed in Appendix A.1, and ViT-L/14 scores, SWD and precision/recall in Appendix Table 8. [†]Pair coherence $\cos(\text{generated image}, \text{generated caption})$, not comparable to the conditional CLIP scores; FID of the joint rows is of the generated image half. [‡]FID is strongly biased in n (+5.3 at $n=5000$ versus 30,000), so these rows are not FID-comparable to CC3M’s (CC3M at $n=5000$: 56.06 / 67.47 at 10 / 256 steps). The last two rows are the floor of the FID column on the same unpaired estimator (Appendix A.1): a perfect model scores 2.35 at this n and 2.94 / 2.55 through the tokenizer, so the codec contributes 0.59 / 0.20.

Comparison to prior work. Table 3 places the result among published zero-shot MS-COCO numbers. Exact comparability is not attainable in this setting and we do not claim it; the table is read for where the model sits, not for rank. Against the one system with the same pre-training corpus and protocol, LAFITE, FID improves by 8.46 without its frozen CLIP encoder. Among unified image+text models we sit between CoDI (400M pairs) and SEED-X (17B parameters), and ahead of the 12B-parameter DALL·E specialist. Every system with a lower FID than ours uses pretrained language or vision components—a text-pretrained backbone or text encoder, a pretrained CLIP or VAE—and most also use substantially more paired data, though several publish no comparable pair count. On GenEval, 0.249 (CC3M) and 0.222 (CC12M) sit above minDALL·E and below CoDI and LlamaGen, far below the LLM-backed unified models. For captioning no zero-shot COCO baseline has been published; COCO-trained specialists reach CIDEr 1.02–1.40 against our 0.136 (*Limitations* below). Taken together, a 462M-parameter model trained from scratch on 2.9M alt-text pairs, with no modality-specific machinery, generates images in the range of unified systems one to two orders of magnitude larger and captions and co-generates from the same weights. That is the claim: capable transfer, not a leaderboard entry.

Sampler: the two modalities want step counts $25\times$ apart. Sweeping the step count from 4 to 512 on the same weights (Appendix A.3), text→image FID and CMMD minimise at 10 steps for CC3M (16–20 for CC12M) and degrade monotonically beyond; at $n=30,000$ the conventional 256 steps costs 4.06 FID for $25.6\times$ the compute. Captioning inverts this: CIDEr is 0.013 at 10 steps, peaks at 0.136 at 256 and falls to 0.103 at 512 (image→text CLIP behaves alike). The joint direction shows both effects inside single samples: every step added improves the text half and degrades the image half (10, 32, 64 steps at $n=5000$: pair-CLIP 23.16, 24.71, 25.17; FID 56.06, 59.14, 62.66), hence the two joint rows of Table 2. This is not a modification of the framework: the step count, like the per-direction churn and guidance (Appendix A.3), is an inference-time hyperparameter of any diffusion sampler, set per direction on the same weights without retraining—and such sampler settings are the only thing that had to be adapted per modality.

Scaling the corpus. Moving to CC12M ($4\times$ the pairs, $1.37\times$ the parameters, $\approx 8.7\times$ the compute, a better codec) improves images and degrades text: text→image FID falls $18.48 \rightarrow 16.59$, a margin

Model	Params	Pre-train pairs	FID-30K↓	GenEval↑
<i>CoBit (ours; from scratch, no pretrained components)</i>				
CoBit-CC3M	462M	2.9M	18.48	0.249
CoBit-CC12M	633M	11.7M	16.59	0.222
<i>Same pre-training corpus (CC3M), zero-shot COCO</i>				
LAFITE (Zhou et al., 2022)	75M (+CLIP)	3.3M	26.94	—
<i>Unified image+text models</i>				
CoDI (Tang et al., 2023a)	—	400M	22.26	0.31
SEED-X (Ge et al., 2024)	17B	—	14.99	0.49
LWM (Liu et al., 2024)	7B	—	12.68	0.47
UniDiffuser (Bao et al., 2023a)	952M	LAION-5B (subsets)	9.71 [†]	—
Show-o (Xie et al., 2024)	1.3B	35M	9.24	0.68
<i>Text-to-image specialists</i>				
DALL·E (Ramesh et al., 2021)	12B	250M	27.50	—
minDALL·E	—	—	—	0.23
LlamaGen	0.8B	—	—	0.32
LDM (Rombach et al., 2022a)	1.4B	400M	12.63	0.37
DALL·E 2 (Ramesh et al., 2022)	6.5B	250M	10.39	0.52
SD v1.5 (Rombach et al., 2022a)	0.9B	2000M	9.62	0.43
PixArt- α (Chen et al., 2024)	0.6B	25M	7.32	0.48
Imagen (Saharia et al., 2022)	3B	860M	7.27	—

Table 3: Zero-shot MS-COCO FID-30K and GenEval overall score against published systems. Prior figures are as printed in the cited papers and in the compilation tables of Xie et al. (2024) (Tables 2–3), Bao et al. (2023a) (Table 1), Zhou et al. (2022) (Table 2) and Ghosh et al. (2023) (Table 2). Rows are not strictly comparable (see text): corpora, backbones, pretrained components and FID pipelines all differ, and the same nominal metric varies by about one FID point between sources (Stable Diffusion is 8.59 in one compilation and 9.62 in another). Parameter counts follow each source’s own convention and do not consistently include frozen components (PixArt- α ’s 0.6B excludes a 4.3B T5 encoder, Imagen’s 3B a frozen T5-XXL, DALL·E 2’s 6.5B includes CLIP), so the column orders magnitude, not cost; DALL·E 2’s prior, decoder and upsamplers saw 250M pairs, not the 650M of its CLIP encoder, and CC3M is 3.3M pairs as released against the 2.9M we could download. CogView (Ding et al., 2021) (blurred, machine-translated MS-COCO, best-of-60 re-ranking) and Improved VQ-Diffusion’s 8.44 (Tang et al., 2022) (COCO-trained) are omitted. [†]Bao et al. (2023a) draw 10K and 30K prompts for FID and CLIP score without saying which is which.

that survives subtracting each tokenizer’s own codec contribution (17.90 versus 16.39; CMMD, 18.41 $\rightarrow$ 17.76, has no such correction and retains an unquantified codec advantage), while at a matched 256 steps captioning CIDEr falls 0.136 $\rightarrow$ 0.057 and image $\rightarrow$ text CLIP 24.83 $\rightarrow$ 22.98. This split is what noisier CC12M alt-text predicts and uniform undertraining does not, but that is an inference from a pattern across metrics, not a measurement; the decisive checkpoint-progression test has not been run (Appendix A.4).

Limitations. Compositional prompt following is near the floor: CC3M scores 0.249 on GenEval (Ghosh et al., 2023) (single object 0.80, colours 0.46, position 0.04, attribute binding 0.03; Appendix A.5) and CC12M 0.222. The score is flat in the step count—0.235 for CC3M at its 10-step image optimum against 0.249 at 256, 0.221 versus 0.222 for CC12M—so this is the model’s compositional ceiling and not a sampler artefact. Captioning is weak in absolute terms (CIDEr 0.136, roughly $7.5\times$ below the COCO-trained L-Verse (Kim et al., 2022) at 1.022; no zero-shot COCO captioning baseline has been published), and the outputs are fluent CC3M-style alt-text scored against COCO’s terse references (RefCLIPScore (Hessel et al., 2021) 0.661; Appendix A.6). Both limitations are consistent with CC3M alt-text rarely specifying relations, attribute bindings or COCO-style descriptions, and both are exactly what the omitted ingredients—text pre-training, curated or re-captioned data, alignment losses—are expected to buy. That is the point of the experiment: it shows what unmodified transfer does and does not provide.

Direction	n	Steps (NFE)	WER↓	CER↓	UTMOS↑	GenPPL-Speech↓
<i>CoBit-LibriTTS — one set of weights (134M, 800 speech tokens)</i>						
text → speech	2,294	512 (1024)	11.7	8.1	3.31	—
speech → text	2,294	512 (1024)	23.3 / 38	15.3 / 24.1	—	—
joint	1,024	256 (256)	20.2	13.2	3.02	395.6
<i>CoBit-MLS — one set of weights (633M, 500 speech tokens)</i>						
text → speech	2,294	512 (1024)	5.3	4.1	4	—
speech → text	2,294	512 (1024)	8.2 / 22	4.3 / 13.1	—	—
joint	1,024	512 (512)	3.7	2.3	3.78	141.2

Table 4: Joint and cross-conditional generative directions from a single set of weights per model for text-audio experiments. Each conditional row is quoted at the best step count for that direction, evaluated on LibriSpeech-PC. Generated audio is transcribed with Whisper-large. Per-direction guidance and churn are listed in Appendix B.1. Text-to-speech evaluation is on test-clean; the two scores for speech-to-text refer to test-clean and test-other, respectively.

Model	Params	UTMOS↑	FSD↓	GenPPL-Speech↓	GPT-4o↑	SALMON↑
<i>CoBit (ours; from scratch, no pretrained components)</i>						
CoBit-LibriTTS	134M	2.78	2,331 / 30.4	150.9	1.11	80.25
CoBit-MLS	633M	3.9	2,183 / 11.9	85.2	1.44	84.25
<i>Trained on MLS-English (same corpus as CoBit-MLS, excluding LibriTTS)</i>						
Flow-SLM-S (Chou et al., 2025)	395M	2.47	3,391 / 35.7	242.8	1.1	82.83
Flow-SLM-L	1.25B	3.48	3,476 / 35.4	131.7	1.57	88.25
<i>Speech-only SLMs</i>						
Flow-SLM-Ext.	1.25B	3.47	3,445.1 / 33	122.4	1.67	90
Llama-Mimi (Sugiura et al., 2025)	1.3B	3.18	3,621 / 14.1	98.5	1.58	87.92
<i>Text+Speech SLMs</i>						
TASLM-Embed (Tseng et al., 2025)	1.3B	4.16	1,410 / 9.9	38.6	3.42	78
SpiritLM-Base (Nguyen et al., 2025)	7B	3.2	7,325 / 26.8	101.4	2	50.5
SpiritLM-Expressive	7B	3.17	6,293.6 / 16.9	153.2	1.62	58.41

Table 5: Continuation-based evaluation for text-audio experiments against published SLMs. All figures are evaluated using publicly available checkpoints, with the exception of generation-based SALMON scores, taken directly from (Hsu et al., 2026). The two FSD records refer to FSD-wlm and FSD-e2v, respectively. More information about FSD and SALMON embeddings is provided on Appendix B.1 We use NFE=256 for both CoBit models.

4.4 AUDIO AND TEXT

Setup. We train two models: CoBit-LibriTTS (134M, 12×768) on LibriTTS (Zen et al., 2019) (354K pairs, 140K steps) and CoBit-MLS (633M, 26×1152) on both LibriTTS and MLS-English (Pratap et al., 2020) (11.2M pairs, 1.2×10^6 steps, 61% of its planned budget). Both datasets consist of single-speaker reading-style paired samples from the LibriVox audiobook corpus. We quantize audio with the FSQ-based StableCodec (Parker et al., 2025) tokenizer using the 1×46656 configuration that produces 25 codes per second (speech budgets of 800 and 500 tokens), prepended by a fixed-length (32 tokens) global speaker representation from BiCodec-global (Wang et al., 2025a). Speaker tokens are not used during waveform decoding; they are included purely as a conditioning signal for generation. Text is tokenized in the same way as in 4.3 (text budget of 168 and 100 tokens); the full text-speaker-speech vocabulary can be captured with 18-bit tokens. Along with unconditional, text-conditional and speech-conditional regimes, training also includes speech continuation, where generation is conditioned on the start (3 seconds) of the audio sample; equal effort (0.25) is allocated to each setting. Evaluation is conducted on LibriSpeech-PC test splits and the SALMON benchmark, neither of which have speaker or example overlaps with the training sets. Text pre-training (Hassid et al., 2023) and speaker conditioning on the audio decoder (Wang et al., 2025a; Tseng et al., 2025) are omitted to demonstrate the generality of the unified bitstream diffusion framework. Both modifications are straightforward in principle: warm-initializing CoBit from a pre-trained text checkpoint mainly required adapting the components that are tied to the sequence length

and the vocabulary bit-width; speaker conditioning for the decoder is simply a matter of fine-tuning. We expect both changes to close the remaining performance gap and leave their implementation to a dedicated line of work on text-audio CoBit.

Results. Joint generation and cross-conditional directions are reported in Table 4, using the same weights for all tasks. CoBit-MLS can co-generate text and audio consistently, achieving a cross-modal WER of 3.7. Character-level disagreement is even lower at 2.3, and a lot of the additional word conflicts can be attributed to the transcription process: the LM-normalized Whisper-large transcriptions of generated audio can have better syntactic structure than the text produced by the diffusion model, which has considerably smaller language modeling capacity. An UTMOS score of 3.78 suggest natural-sounding audio, even when sampling the speaker identity from scratch as well. The content also remains highly plausible in terms of perplexity under an external LM, despite generation without a text or speech prompt. Both CoBit versions are considerably stronger in text-to-speech synthesis than in speech recognition. In speech continuation, shown in Table 5, both models sample acoustically and linguistically grounded audio, yet they struggle to preserve thematic content. *[Joint generation results use stochasticity; verify this is not an issue.]*

Comparison to prior work. Table 5 places speech continuation and generation-based SALMON (Hsu et al., 2026) performance among autoregressive speech language models; baseline comparisons for co-generation and cross-modal transfer are offered in Tables 17 and 18, respectively. Per-task SALMON scores are provided in Table 19. Further specification about the comparability of these metrics is provided in Appendix B.6. The capabilities of the bitstream diffusion framework are best shown in speech continuation, where CoBit-MLS outperforms all baselines except TASLM-Embed on UTMOS, FSD-wlm, FSD-e2v and GenPPL. Even CoBit-LibriTTS scores better than the smallest autoregressive model (Flow-SLM-S) across these metrics and matches its weak LLM-as-a-Judge performance, despite having less than half the parameters and only a fraction of the training data. LLM-as-a-Judge is in general CoBit’s weakest metric; its fixed-rate speech tokenization and from-scratch initialization limit its ability to produce thematically relevant continuations. Even warm initialization, however, appears to be insufficient for semantic extrapolation; thematically accurate speech is only achieved by TASLM-Embed, which uses text-aligned speech representations that close the semantic gap between text and speech modelling. In SALMON-based evaluation, CoBit scores better on average than all text-speech SLMs but cannot match speech-only models. While CoBit remains competitive on voice-related tasks with the aid of its speaker features, it is worse than Flow-SLM and Llama-Mimi models on noise-related tasks; this can be attributed to the Mimi tokenizer (Défossez et al., 2024) they use, which captures more fine-grained details of the audio signal. In text-to-speech, CoBit-MLS surpasses SpiritLM on CER (4.1 to 6.7) and approaches Moshi on WER (5.3 to 4.7), despite being 10 times smaller. However, despite the consistency and naturalness of generated audio, speaker characteristics are not reproduced to the same level as by the ASR-TTS hybrid GPA-v1.5 (Cai et al., 2026) when the decoder is not conditioned on the speaker. Speech recognition is where the absence of text initialization is the most apparent, as it contradicts the design of ASR-specific systems, which typically recognize speech at the character or phoneme level using self-supervised representations, and then employ a text LM to decode their output into well-structured sentences (Radford et al., 2023). For SLMs, the textually pre-trained backbone implicitly plays a similar role, leading to more coherent transcriptions, as previously demonstrated (Maiti et al., 2024).

Sampler. *[Decide what settings need to be considered to match image/biological experiments]*

Effect of model and corpus scale. Increasing the model scale and adding MLS-English to the training set improves performance across all tasks and metrics. MLS-English is much larger (45k hours compared to 580), preserves reading-style audio and is part of StableCodec’s training corpus as well, eliminating the risk of distributional shifts during tokenization. CoBit-MLS is better at aligning text and speech sequences when generating both from scratch, yet CoBit-LibriTTS achieves the best error rates when given the text as input. The performance gap is the smallest for SALMON accuracy, as the smaller model can still capture acoustic properties of the speech prompt. Despite its linguistic fluency and frame-level realism, the main limitation of CoBit-LibriTTS is producing natural-sounding audio, observed across all speech-emitting tasks in terms of UTMOS.

Limitations. Across both models, LLM-as-a-Judge is often at the lowest possible score (Figure X) with averages of 1.11 and 1.44 for CoBit-LibriTTS and CoBit-MLS, respectively. It is very difficult to infer relevant continuations to a speech prompt when the training data is semantically limited

to storybook passages. Speech-to-text transfer is also weak, but a large portion of the errors are related only to word-level alignment (WER is almost $\times 2$ CER on test-clean for CoBit-MLS). These limitations are strongly tied to the lack of text pre-training. Imprecise speaker replication in terms of SpkSim can be directly associated with the speaker features being used only in the diffusion process and not in the audio decoder. While all three issues could be accounted for by adjustments that are well established in speech language modeling literature, this work demonstrates what the generalized bitstream diffusion framework can achieve off-the-shelf.

4.5 PROTEIN SEQUENCE AND STRUCTURE

Unlike text, image, and audio multi-modality, whose correspondence is primarily semantic, a protein sequence and its three-dimensional structure are physically coupled descriptions. We ask whether the bitstream formulation can learn the coupled modalities of protein sequence and structure by training a 396M-parameter denoiser from scratch on 216,237 PDB/AFDB-SwissProt (Berman et al., 2000; Varadi et al., 2024) pairs with 20% sequence replay. Each residue aligns five amino-acids bits with thirteen signs from the frozen DPLM-2 lookup-free quantiser (Wang et al., 2024b). Decoding retains the 20 canonical amino acids and all 2^{13} structural codes. The common binary interface lets one denoiser learn sequence-structure dependencies while the frozen codec recovers backbone coordinates. Data, splits, and tokenisation are detailed in Appendices • Bruno: ref appendix.

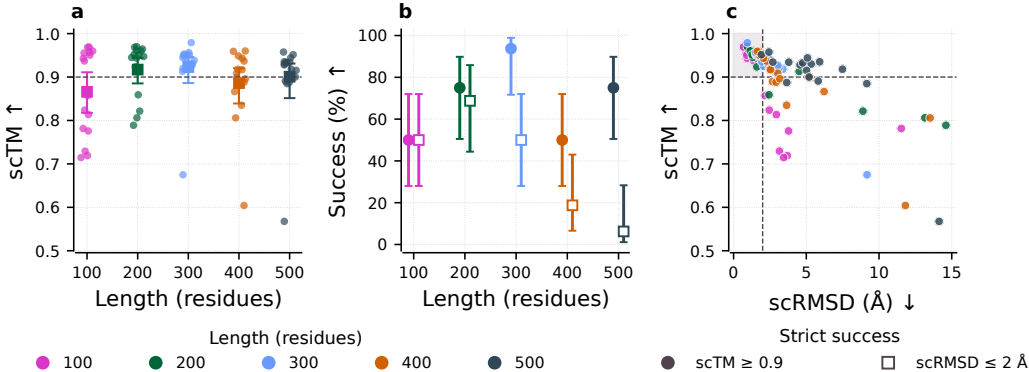


Figure 1: Backbone quality from denoised bits: (a) scTM by length, (b) strict success by length, and (c) fold similarity versus coordinate error. Colours denote length. 80 backbones each receive one ProteinMPNN redesign and ESMFold refold. Intervals are 95%. Shading marks both strict criteria.

Tasks coverage. Clamping either channel gives inverse or forwarding folding while leaving both free engenders joint generation, without task-specific fine-tuning or Multiple Sequence Alignment as input • Bruno: cite MSA?. Our evaluations were based on entropic DDIM with 50 grid points with conditional CFG 5 and no unconditional guidance (R). Under these settings, mean recovery is 44.10% on 128 CATH 4.3 validation targets and forward TM is .554 on 183 CAMEO Targets (Sillitoe et al., 2021; Haas et al., 2018; Zhang & Skolnick, 2004). These results test bidirectional learning through one shared bitstream objective. For 80 generated backbones (16 per length, 100–500), one ProteinMPNN redesign and ESMFold refold give mean scTM .899 (Dauparas et al., 2022; Lin et al., 2023). Figure 1 shows fold similarity across lengths while Figure 2 shows the best backbones per length-bucket. We assess backbone recovery after redesign and refolding using $\text{scTM} \geq .9$ (D_{TM}) for overall fold agreement and $\text{scRMSD} \leq 2 \text{ \AA}$ (D_{RMSD}) for close coordinate agreement. Full metrics and sampling analysis are in Appendices • Bruno: ?.

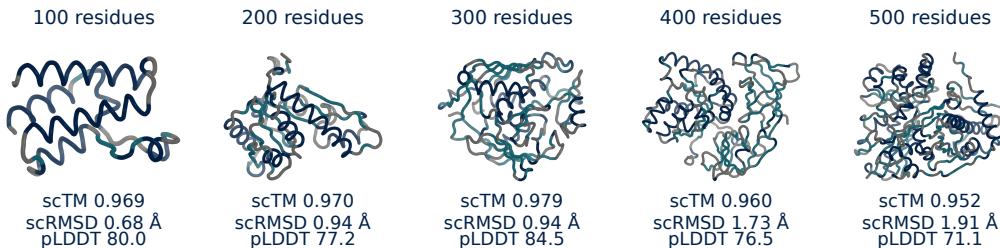


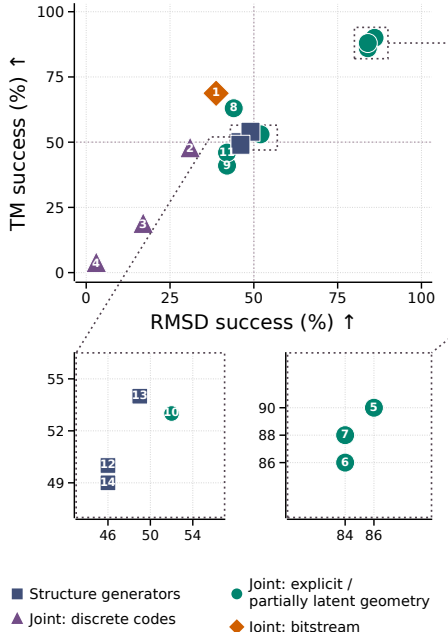
Figure 2: Selected CoBit backbones, rendered with PyMOL (Schrödinger, LLC, 2026). Highest scTM at each length among samples with scRMSD ≤ 2 Å; scores use ProteinMPNN/ESMFold. pLDDT is predicted confidence.

Bitstream capability versus protein optimisation. We compare CoBit’s bitstream formulation against protein-specific models. Table 6 inserts CoBit into the extensive research done by (Lu et al., 2026). MultiFlow Campbell et al. (2024) leads on the strict criteria with a 21.8M-parameters model due to its explicit geometric learning and redesign-select and refolding-filtered training. On the other hand, CoBit tests the common bit objective with an existing codec and paired-data recipe. Its sequence replay supplies unpaired example unlike MultiFlow’s designability distillation or direct geometric supervision. The unmatched comparison cannot isolate representation of parameter efficiency. • Bruno: Appendix-X explains the training differences and MultiFlow’s distillation ablation.

Table 6: Backbone designability (%) with one ProteinMPNN redesign and ESMFold. Published baselines: Lu et al. (2026); G: general bitstream framework; P: protein-specific generator.

ID	Model	Scope	n	D_{RMSD}	D_{TM}
<i>Our evaluation: joint bitstream generator</i>					
1	CoBit	G	80	38.8	68.8
<i>Published evaluation: Lu et al. (2026)</i>					
Joint generators: discrete structural codes					
2	DPLM-2 (650M)	P	100	31.0	48.0
3	ESM3 Open (sequence first)	P	100	17.0	19.0
4	ESM3 Open (structure first)	P	100	3.0	4.0
Joint generators: explicit / partially latent geometry					
5	MultiFlow	P	100	86.0	90.0
6	La-Proteina (no-triangle)	P	100	84.0	86.0
7	La-Proteina (triangle)	P	100	84.0	88.0
8	SimpleDesign	P	100	44.0	63.0
9	ProtPardelle	P	100	42.0	41.0
10	ProtPardelle-1c	P	100	52.0	53.0
11	ProteinGenerator	P	100	42.0	46.0
Structure generators					
12	Proteina	P	100	46.0	50.0
13	RFdiffusion	P	100	49.0	54.0
14	FrameFlow	P	100	46.0	49.0

Figure 3: Visual representation of the rates from Table 6. CoBit captures protein organisation better than its precise backbone geometry.



A dedicated protein study can test that coupling, stronger supervision, and scaling can be optimised for biological tasks such as binder-linker design with laboratory (wet lab) validation. CoBit’s stronger fold-level (D_{TM}) than coordinate-level (D_{RMSD}) success indicates that denoising structural bits capture useful protein organisation information and that geometric precision can be further leveraged in a dedicated protein-training work. The present metrics indicate modeling quality, but biological function requires those improvements and additional experiments.

5 DISCUSSION

[To be written.]

6 CONCLUSION

[To be written.]

REFERENCES

- Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, Colin Raffel, Leandro von Werra, and Thomas Wolf. Smollm2: When smol goes big – data-centric training of a small language model, 2025. URL <https://arxiv.org/abs/2502.02737>.
- Peter Anderson, Basura Fernando, Mark Johnson, and Stephen Gould. Spice: Semantic propositional image caption evaluation. In *European Conference on Computer Vision (ECCV)*, 2016.
- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems*, 2021. URL <https://arxiv.org/abs/2107.03006>.
- Fan Bao, Shen Nie, Kaiwen Xue, Yue Cao, Chongxuan Li, Hang Su, and Jun Zhu. One transformer fits all distributions in multi-modal diffusion at scale. In *International Conference on Machine Learning (ICML)*, 2023a.
- Fan Bao, Shen Nie, Kaiwen Xue, Chongxuan Li, Shi Pu, Yaole Wang, Gang Yue, Yue Cao, Hang Su, and Jun Zhu. One transformer fits all distributions in multi-modal diffusion at scale. In *Proceedings of the 40th International Conference on Machine Learning*, 2023b. URL <https://proceedings.mlr.press/v202/bao23a.html>.
- Georgios Batzolis, Mark Girolami, and Luca Ambrogioni. CoBit: Language modeling with bit-stream diffusion, 2026a. URL <https://arxiv.org/abs/2605.07013>.
- Georgios Batzolis, Mark Girolami, and Luca Ambrogioni. Cobit: Language modeling with bit-stream diffusion. *arXiv preprint arXiv:2605.07013*, 2026b.
- Helen M. Berman, John Westbrook, Zukang Feng, Gary Gilliland, T. N. Bhat, Helge Weissig, Ilya N. Shindyalov, and Philip E. Bourne. The protein data bank. *Nucleic Acids Research*, 28(1):235–242, 2000. doi: 10.1093/nar/28.1.235.
- Runyuan Cai, Yu Lin, Yiming Wang, Chunlin Fu, and Xiaodong Zeng. Unifying speech recognition, synthesis and conversion with autoregressive transformers, 2026. URL <https://arxiv.org/abs/2601.10770>.
- Andrew Campbell, Jason Yim, Regina Barzilay, Tom Rainforth, and Tommi Jaakkola. Generative flows on discrete state-spaces: Enabling multimodal flows with applications to protein co-design. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. URL <https://arxiv.org/abs/2402.04997>.
- Soravit Changpinyo, Piyush Sharma, Nan Ding, and Radu Soricut. Conceptual 12m: Pushing web-scale image-text pre-training to recognize long-tail visual concepts. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- Junsong Chen, Jincheng Yu, Chongjian Ge, Lewei Yao, Enze Xie, Yue Wu, Zhongdao Wang, James Kwok, Ping Luo, Huchuan Lu, and Zhenguo Li. Pixart- α : Fast training of diffusion transformer for photorealistic text-to-image synthesis. In *International Conference on Learning Representations (ICLR)*, 2024.

- Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, et al. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518, 2022.
- Ting Chen, Ruixiang Zhang, and Geoffrey Hinton. Analog bits: Generating discrete data using diffusion models with self-conditioning. In *International Conference on Learning Representations*, 2023a. URL <https://openreview.net/forum?id=3itjR9QxFw>.
- Ting Chen, Ruixiang Zhang, and Geoffrey Hinton. Analog bits: Generating discrete data using diffusion models with self-conditioning. In *International Conference on Learning Representations (ICLR)*, 2023b.
- Yuxin Chen, Chumeng Liang, Hangke Sui, Ruihan Guo, Chaoran Cheng, Jiaxuan You, and Ge Liu. LangFlow: Continuous diffusion rivals discrete in language modeling, 2026. URL <https://arxiv.org/abs/2604.11748>.
- Ju-Chieh Chou, Jiawei Zhou, and Karen Livescu. Flow-slm: Joint learning of linguistic and acoustic information for spoken language modeling. *arXiv preprint arXiv:2508.09350*, 2025.
- Marcella Cornia, Matteo Stefanini, Lorenzo Baraldi, and Rita Cucchiara. Meshed-memory transformer for image captioning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- Justas Dauparas, Ivan Anishchenko, Nathaniel Bennett, Hua Bai, Robert J. Ragotte, Lukas F. Milles, Basile I. M. Wicky, Alexis Courbet, Rob J. de Haas, Neville Bethel, et al. Robust deep learning-based protein sequence design using ProteinMPNN. *Science*, 378(6615):49–56, 2022. doi: 10.1126/science.add2187.
- Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- Brecht Desplanques, Jenthe Thienpondt, and Kris Demuynck. Ecapa-tdnn: Emphasized channel attention, propagation and aggregation in tdnn based speaker verification. *arXiv preprint arXiv:2005.07143*, 2020.
- Sander Dieleman, Laurent Sartran, Arman Roshannai, Nikolay Savinov, Yaroslav Ganin, Pierre H. Richemond, Arnaud Doucet, Robin Strudel, Chris Dyer, Conor Durkan, Curtis Hawthorne, Rémi Leblond, Will Grathwohl, and Jonas Adler. Continuous diffusion for categorical data, 2022. URL <https://arxiv.org/abs/2211.15089>.
- Ming Ding, Zhuoyi Yang, Wenyi Hong, Wendi Zheng, Chang Zhou, Da Yin, Junyang Lin, Xu Zou, Zhou Shao, Hongxia Yang, and Jie Tang. Cogview: Mastering text-to-image generation via transformers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Yuying Ge, Sijie Zhao, Jinguo Zhu, Yixiao Ge, Kun Yi, Lin Song, Chen Li, Xiaohan Ding, and Ying Shan. Seed-x: Multimodal models with unified multi-granularity comprehension and generation. *arXiv preprint arXiv:2404.14396*, 2024.
- Dhruba Ghosh, Hannaneh Hajishirzi, and Ludwig Schmidt. Geneval: An object-focused framework for evaluating text-to-image alignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Jürgen Haas, Alessandro Barbato, Dario Behringer, Gabriel Studer, Steven Roth, Martino Bertoni, Khaled Mostaguir, Rafal Gumienny, and Torsten Schwede. Continuous automated model Evaluation (CAMEO) complementing the critical assessment of structure prediction in CASP12. *Proteins: Structure, Function, and Bioinformatics*, 86(Suppl. 1):387–398, 2018. doi: 10.1002/prot.25431. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC5820194/>.
- Michael Hassid, Tal Remez, Tu Anh Nguyen, Itai Gat, Alexis Conneau, Felix Kreuk, Jade Copet, Alexandre Defossez, Gabriel Synnaeve, Emmanuel Dupoux, et al. Textually pretrained speech language models. *Advances in Neural Information Processing Systems*, 36:63483–63501, 2023.

- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *Empirical Methods in Natural Language Processing (EMNLP)*, 2021.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, 2021.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2006.11239>.
- Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J. Fleet. Video diffusion models. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2204.03458>.
- Chan-Jan Hsu, Liang-Hsuan Tseng, Yi-Cheng Lin, Yen-Chun Kuo, Ju-Chieh Chou, Kai-Wei Chang, Hung-yi Lee, and Carlos Busso. On the fallacy of global token perplexity in spoken language model evaluation. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 37000–37020, 2026.
- Keya Hu, Linlu Qiu, Yiyang Lu, Hanhong Zhao, Tianhong Li, Yoon Kim, Jacob Andreas, and Kaiming He. ELF: Embedded language flows, 2026. URL <https://arxiv.org/abs/2605.10938>.
- Sadeep Jayasumana, Srikumar Ramalingam, Andreas Veit, Daniel Glasner, Ayan Chakrabarti, and Sanjiv Kumar. Rethinking fid: Towards a better evaluation metric for image generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Taehoon Kim, Gwangmo Song, Sihaeng Lee, Sangyun Kim, Yewon Seo, Soonyoung Lee, Seung Hwan Kim, Honglak Lee, and Kyunghoon Bae. L-verse: Bidirectional generation between image and text. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Nithin Rao Koluguri, Taejin Park, and Boris Ginsburg. Titanet: Neural model for speaker representation with 1d depth-wise separable convolutions and global context. In *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pp. 8102–8106. IEEE, 2022.
- Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. DiffWave: A versatile diffusion model for audio synthesis. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2009.09761>.
- Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Improved precision and recall metric for assessing generative models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Moreno La Quatra, Alkis Koudounas, Lorenzo Vaiani, Elena Baralis, Luca Cagliero, Paolo Garza, and Sabato Marco Siniscalchi. Benchmarking representations for speech, music, and acoustic events. In *2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops (ICASSPW)*, pp. 505–509, 2024. doi: 10.1109/ICASSPW62465.2024.10625960.
- Xiang Lisa Li, John Thickstun, Ishaan Gulrajani, Percy Liang, and Tatsunori B. Hashimoto. Diffusion-LM improves controllable text generation. In *Advances in Neural Information Processing Systems*, 2022. URL <https://arxiv.org/abs/2205.14217>.

- Xiujun Li, Xi Yin, Chunyuan Li, Pengchuan Zhang, Xiaowei Hu, Lei Zhang, Lijuan Wang, Houdong Hu, Li Dong, Furu Wei, Yejin Choi, and Jianfeng Gao. Oscar: Object-semantics aligned pre-training for vision-language tasks. In *European Conference on Computer Vision (ECCV)*, 2020.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C. Lawrence Zitnick. Microsoft coco: Common objects in context. In *European Conference on Computer Vision (ECCV)*, 2014.
- Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379:1123–1130, 2023. doi: 10.1126/science.ade2574.
- Hao Liu, Wilson Yan, Matei Zaharia, and Pieter Abbeel. World model on million-length video and language with blockwise ringattention. *arXiv preprint arXiv:2402.08268*, 2024.
- Jiarui Lu, Yuyang Wang, Yizhe Zhang, Jiatao Gu, Navdeep Jaitly, Joshua M. Susskind, and Miguel Ángel Bautista. SimpleDesign: A joint model for protein sequence and structure code-sign. *Transactions on Machine Learning Research*, 2026. URL <https://arxiv.org/abs/2609.03377>.
- Zhuoyan Luo, Fengyuan Shi, Yixiao Ge, Yujiu Yang, Limin Wang, and Ying Shan. Open-magvit2: An open-source project toward democratizing auto-regressive visual generation. In *arXiv preprint arXiv:2409.04410*, 2024.
- Ziyang Ma, Zhisheng Zheng, Jiaxin Ye, Jinchao Li, Zhifu Gao, Shiliang Zhang, and Xie Chen. emotion2vec: Self-supervised pre-training for speech emotion representation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 15747–15760, 2024.
- Soumi Maiti, Yifan Peng, Shukjae Choi, Jee-weon Jung, Xuankai Chang, and Shinji Watanabe. VoxTlm: Unified decoder-only models for consolidating speech recognition, synthesis and speech, text continuation tasks. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 13326–13330. IEEE, 2024.
- Fabian Mentzer, David Minnen, Eirikur Agustsson, and Michael Tschannen. Finite scalar quantization: Vq-vae made simple. In *International Conference on Learning Representations*, volume 2024, pp. 51772–51783, 2024.
- Tu Anh Nguyen, Benjamin Muller, Bokai Yu, Marta R Costa-Jussa, Maha Elbayad, Sravya Popuri, Christophe Ropers, Paul-Ambroise Duquenne, Robin Algayres, Ruslan Mavlyutov, et al. Spirit-lm: Interleaved spoken and written language model. *Transactions of the Association for Computational Linguistics*, 13:30–52, 2025.
- Julian Parker, Anton Smirnov, Jordi Pons, CJ Carr, Zack Zukowski, Zach Evans, and Xubo Liu. Scaling transformers for low-bitrate high-quality speech coding. In *International Conference on Learning Representations*, volume 2025, pp. 51997–52021, 2025.
- Vineel Pratap, Qiantong Xu, Anuroop Sriram, Gabriel Synnaeve, and Ronan Collobert. Mls: A large-scale multilingual dataset for speech research. *arXiv preprint arXiv:2012.03411*, 2020.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pp. 28492–28518. PMLR, 2023.
- Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. Zero-shot text-to-image generation. In *International Conference on Machine Learning (ICML)*, 2021.
- Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125*, 2022.

- Gabriel Raya, Bac Nguyen, Georgios Batzolis, Yuhta Takida, Dejan Stancevic, Naoki Murata, Chieh-Hsin Lai, Yuki Mitsufuji, and Luca Ambrogioni. Noise scheduling as information-guided allocation in diffusion training, 2026. URL <https://arxiv.org/abs/2602.18647>.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022a.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022b. URL https://openaccess.thecvf.com/content/CVPR2022/html/Rombach_High-Resolution_Image_Synthesis_With_Latent_Diffusion_Models_CVPR_2022_paper.html.
- Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily L. Denton, Kamyar Ghasemipour, Raphael Gontijo Lopes, Burcu Karagol Ayan, Tim Salimans, Jonathan Ho, David J. Fleet, and Mohammad Norouzi. Photorealistic text-to-image diffusion models with deep language understanding. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Subham S. Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T. Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In *Advances in Neural Information Processing Systems*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/eb0b13cc515724ab8015bc978fdde0ad-Abstract-Conference.html.
- Schrödinger, LLC. The PyMOL molecular graphics system. Version 3.2, open-source build, 2026. URL <https://pymol.org/>.
- Piyush Sharma, Nan Ding, Sebastian Goodman, and Radu Soricut. Conceptual captions: A cleaned, hypernamed, image alt-text dataset for automatic image captioning. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018.
- Ian Sillitoe, Nicola Bordin, Natalie Dawson, Vaishali P. Waman, Paul Ashford, Harry M. Scholes, Camilla S. M. Pang, Laurel Woodridge, Clemens Rauer, Neeladri Sen, et al. CATH: Increased structural coverage of functional space. *Nucleic Acids Research*, 49(D1):D266–D273, 2021. doi: 10.1093/nar/gkaa1079. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC7778904/>.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations (ICLR)*, 2021a.
- Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021b. URL <https://arxiv.org/abs/2011.13456>.
- Dejan Stancevic, Florian Handke, and Luca Ambrogioni. Entropic time schedulers for generative diffusion models. In *Advances in Neural Information Processing Systems*, 2025. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/3ed10616ecdd776be283c0a45cf9332d-Abstract-Conference.html.
- Issa Sugiura, Shuhei Kurita, Yusuke Oda, and Ryuichiro Higashinaka. Llama-mimi: Speech language models with interleaved semantic and acoustic tokens. *arXiv preprint arXiv:2509.14882*, 2025.
- Zhicong Tang, Shuyang Gu, Jianmin Bao, Dong Chen, and Fang Wen. Improved vector quantized diffusion models. *arXiv preprint arXiv:2205.16007*, 2022.
- Zineng Tang, Ziyi Yang, Chenguang Zhu, Michael Zeng, and Mohit Bansal. Any-to-any generation via composable diffusion. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023a.

- Zineng Tang, Ziyi Yang, Chenguang Zhu, Michael Zeng, and Mohit Bansal. Any-to-any generation via composable diffusion. In *Advances in Neural Information Processing Systems*, 2023b. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/33edf072fe44f19079d66713a1831550-Abstract-Conference.html.
- Lucas Theis, Aäron van den Oord, and Matthias Bethge. A note on the evaluation of generative models. *arXiv preprint arXiv:1511.01844*, 2015.
- Liang-Hsuan Tseng, Yi-Chang Chen, Kuan-Yi Lee, Da-Shan Shiu, and Hung-yi Lee. Taste: Text-aligned speech tokenization and embedding for spoken language modeling. *arXiv preprint arXiv:2504.07053*, 2025.
- Mihaly Varadi, Damian Bertoni, Paulyna Magana, et al. AlphaFold protein structure database in 2024: providing structure coverage for over 214 million protein sequences. *Nucleic Acids Research*, 52(D1):D368–D375, 2024. doi: 10.1093/nar/gkad1011.
- Xinsheng Wang, Mingqi Jiang, Ziyang Ma, Ziyu Zhang, Songxiang Liu, Linqin Li, Zheng Liang, Qixi Zheng, Rui Wang, Xiaoqin Feng, et al. Spark-tts: An efficient llm-based text-to-speech model with single-stream decoupled speech tokens. *arXiv preprint arXiv:2503.01710*, 2025a.
- Xinyou Wang, Zaixiang Zheng, Fei Ye, Dongyu Xue, Shujian Huang, and Quanquan Gu. DPLM-2: A multimodal diffusion protein language model. *arXiv preprint arXiv:2410.13782*, 2024a. URL <https://arxiv.org/abs/2410.13782>.
- Xinyou Wang, Zaixiang Zheng, Fei Ye, Dongyu Xue, Shujian Huang, and Quanquan Gu. Diffusion language models are versatile protein learners. In *Proceedings of the 41st International Conference on Machine Learning*, 2024b. URL <https://proceedings.mlr.press/v235/wang24ct.html>.
- Xinyou Wang, Zaixiang Zheng, Fei Ye, Dongyu Xue, Shujian Huang, and Quanquan Gu. DPLM-2: A multimodal diffusion protein language model. In *International Conference on Learning Representations*, 2025b. URL <https://openreview.net/forum?id=5z9GjHgerY>.
- Jinheng Xie, Weijia Mao, Zechen Bai, David Junhao Zhang, Weihao Wang, Kevin Qinghong Lin, Yuchao Gu, Zhijie Chen, Zhenheng Yang, and Mike Zheng Shou. Show-o: One single transformer to unify multimodal understanding and generation. *arXiv preprint arXiv:2408.12528*, 2024.
- Lijun Yu, Jose Lezama, Nitesh B. Gundavarapu, Luca Versari, Kihyuk Sohn, David Minnen, Yong Cheng, Agrim Gupta, Xiuye Gu, Alexander G. Hauptmann, Boqing Gong, Ming-Hsuan Yang, Irfan Essa, David A. Ross, and Lu Jiang. Language model beats diffusion – tokenizer is key to visual generation. In *International Conference on Learning Representations (ICLR)*, 2024.
- Heiga Zen, Viet Dang, Rob Clark, Yu Zhang, Ron J Weiss, Ye Jia, Zhifeng Chen, and Yonghui Wu. Libritts: A corpus derived from librispeech for text-to-speech. *arXiv preprint arXiv:1904.02882*, 2019.
- Yang Zhang and Jeffrey Skolnick. Scoring function for automated assessment of protein structure template quality. *Proteins: Structure, Function, and Bioinformatics*, 57(4):702–710, 2004. doi: 10.1002/prot.20264. URL <https://pubmed.ncbi.nlm.nih.gov/15476259/>.
- Yufan Zhou, Ruiyi Zhang, Changyou Chen, Chunyuan Li, Chris Tensmeyer, Tong Yu, Jiuxiang Gu, Jinhui Xu, and Tong Sun. Lafite: Towards language-free training for text-to-image generation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.

A IMAGE AND TEXT EXPERIMENTS

Supporting material for §4.3. Every figure was read from the job log of the run that produced it. Unless stated otherwise, sampling is deterministic DDIM on the entropy-derived grid with `carry` self-conditioning and $\sigma_{\text{dec}}=0.1$.

	CoBit-CC3M	CoBit-CC12M
Training corpus	CC3M (≈ 2.9 M pairs)	CC12M (≈ 11.7 M pairs)
Parameters; layers $\times$ width	462M; 24×1024	633M; 26×1152
Image tokenizer (Open-MAGVIT2, 18-bit LFQ)	$f16$: $16 \times 16 = 256$	$f8$: $32 \times 32 = 1,024$
Text budget L_t ; positions; bits S	48; 310; 5,890	96; 1,126; 21,394
Training steps	1,000,000	1,744,000 (87% of target)
<i>Relative training compute</i>	$1 \times$	$\approx 8.7 \times$
<i>Tokenizer rFID, paired (COCO-30K)</i>	1.592	0.726
<i>Codec contribution to an unpaired FID</i>	0.588	0.199
<i>Tokenizer PSNR (COCO-30K)</i>	22.24 dB	26.92 dB

Table 7: The two checkpoints. CC12M has $4 \times$ the data, $1.37 \times$ the parameters, $3.63 \times$ the positions per step and $1.74 \times$ the steps ($\approx 8.7 \times$ the compute), and a materially better codec, so every image-side comparison in §A.4 is biased in CC12M’s favour. The paired rFID is the Fréchet distance between the 30,000 real images and their own tokenizer round-trip; the row below it is the codec’s contribution when the two sets are disjoint, which is the quantity comparable to a generative FID (§A.1).

A.1 PROTOCOL AND CHECKPOINTS

Zero-shot MS-COCO. 30,000 captions are sampled with a fixed seed from COCO val2014 (Lin et al., 2014), one image is generated per caption, and FID (Heusel et al., 2017) is computed with Inception-V3 pool3 features against the 30,000 real images (both resized on the shorter side to 256 and centre-cropped). CLIP score is $100 \cdot \overline{\cos}(\text{image}, \text{text})$ with OpenAI ViT-B/32 (Radford et al., 2021); ViT-L/14 is given in Table 8 because published CLIP scores use that backbone and the two scales differ substantially. CMMD (Jayasumana et al., 2024) and SWD are computed in CLIP ViT-B/32 space against a 30,000-image real bank; precision and recall follow Kynkäänniemi et al. (2019) in the same space ($k=3$). Captioning uses the complete Karpathy test split (5,000 images) with `pycocoevalcap` including SPICE (Anderson et al., 2016); CLIPScore and RefCLIPScore follow Hessel et al. (2021). GenEval (Ghosh et al., 2023) uses the official harness (553 prompts $\times$ 4 images). The FID reference statistics and prompt captions are byte-identical between the $f16$ and $f8$ asset builds, so the text $\rightarrow$ image comparison between the two models is exact on the image side.

Operating points. Classifier-free guidance evaluates the conditional and null branches as one batch (§3.3), so $\text{NFE} = 2 \times \text{steps}$ for guided directions and $= \text{steps}$ for joint generation. Text $\rightarrow$ image: deterministic, guidance 2.5 (CC3M) / 2.0 (CC12M), 10 / 16 steps. Image $\rightarrow$ text CLIP: churn $\gamma=0.175$, guidance 3.0, 256 steps. Captioning: churn $\gamma=0.175$, guidance 2.0, 256 steps. Joint: deterministic, unguided. Each was chosen by a bracketed sweep at $n=5000$ read on CMMD (CIDEr for captioning) and confirmed at the full protocol where one exists.

Sample-count bias. FID is strongly biased in n : at a fixed configuration CC3M’s FID is 28.10 at $n=5000$ and 22.79 at $n=30,000$ (256 steps), and 24.00 versus 18.73 at 10 steps, so the +5.3 bias is a property of n alone; CMMD moves by under one point ($22.57 \rightarrow 22.16$). Sweeps are therefore read on CMMD, FID is quoted at $n=30,000$ wherever a full-protocol run exists, and FID is never compared across different n .

The floor of the FID column. FID at this n is also bounded away from zero: a disjoint 30,000-image real draw from COCO train2014 (seed 42, zero image-id overlap; neither model trains on COCO) scores **2.35** against the same reference, 99% of it in the covariance term—finite-sample estimation, not a train/val shift, which would show in the mean. The same draw after a tokenizer round-trip gives 2.94 ($f16$) and 2.55 ($f8$), so the codec adds 0.59 and 0.20, and those are the numbers a generative FID is read against here. The paired reconstruction FID of Table 7 ($1.59 / 0.73$) compares images with their own round-trip and so omits the finite-sample term that every disjoint-set FID carries, the generative one included; it is a codec statistic, not a quantity to subtract from a generative FID.

Direction	n	gs	Steps	FID↓	CMMD↓	SWD↓	CLIP↑		Prec.	Rec.
							B/32	L/14		
<i>CoBit-CC3M</i>										
text → image	30,000	2.5	10	18.48	18.41	0.632	27.73	—	—	—
text → image	30,000	3.0	10	18.73	18.56	0.631	28.03	21.66	0.48	0.28
text → image	30,000	3.0	256	22.79	22.16	0.667	28.02	21.79	0.41	0.29
image → text	10,000	3.0	256	—	—	—	24.86	18.30	—	—
image → text	5,000	3.0	256	—	—	—	24.83	—	—	—
image → text	10,000	4.0	256	—	—	—	24.99	—	—	—
joint (unconditional)	30,000	—	10	50.19	41.53	0.904	23.10	16.87	0.36	0.22
joint (unconditional)	30,000	—	256	61.95	46.80	0.971	25.67	—	0.20	0.30
<i>CoBit-CC12M</i>										
text → image	30,000	2.0	16	16.59	17.76	0.591	26.51	—	—	—
text → image	30,000	2.0	20	16.70	17.68	0.585	26.54	—	—	—
image → text	5,000	3.0	256	—	—	—	22.98	—	—	—
image → text	5,000	4.0	256	—	—	—	23.08	—	—	—
joint (unconditional)	5,000	—	20	56.18	37.60	0.837	23.19	—	—	—
joint (unconditional)	5,000	—	256	64.47	40.33	0.878	24.01	—	—	—
<i>FID floor, sampling</i>	30,000			2.35	—	—	—	—	—	—
<i>FID floor, f16 / f8</i>	30,000			2.94 / 2.55	—	—	—	—	—	—

Table 8: Full version of Table 2. The CLIP column of the joint rows is pair coherence between the generated image and caption. Guidance 2.5 is marginally better than 3.0 for CC3M text→image on FID and CMMD and 0.30 worse on CLIP; L/14, precision and recall were measured at guidance 3.0 only. Moving CC3M from 256 to 10 steps raises precision from 0.41 to 0.48 with recall 0.29→0.28, so the gain is not a diversity-for-fidelity trade. On the L/14 column UniDiffuser (Bao et al., 2023a) reports roughly 25–26 (text→image) and 22 (image→text) against our 21.66–21.79 and 18.30, consistent with the FID gap and its LAION-scale pre-training. Two repeats of the 256-step headline five days apart, at world sizes 2 and 4, gave FID 22.7843 and 22.7881 (CMMD 22.1704 / 22.1627); they bound seed and sharding variation together, so the ≈ 0.05 FID threshold they set at $n=30,000$ is a conservative one. The last two rows are the floor of the FID column (§A.1): 2.35 from finite-sample estimation alone, 2.94 / 2.55 once the tokenizer round-trip is included.

A.2 COMPLETE RESULTS

Table 8 extends Table 2 with SWD, the ViT-L/14 CLIP backbone, precision and recall, the alternative guidance and step settings, and the superseded 256-step text→image operating point.

A.3 SAMPLER CHARACTERISATION

Tables 9–11 give the step-count and guidance sweeps behind the operating points of §A.1.

Reading. The image optimum is about 10 steps and the text optimum about 256, a factor of 25, in a model that makes no architectural distinction between the two token populations. There is therefore no single operating point for a joint model, and any single-number report of one is a choice about which modality to favour; we quote each direction at its own optimum and say so in every table. Since architecture, weights, bitstream and σ grid are shared, the asymmetry must be a property of the token statistics. A plausible account, which we have not tested, is that image tokens in an 18-bit LFQ code are locally redundant and strongly correlated across the grid, so a coarse trajectory lands in the right basin and additional steps mainly accumulate and re-inject self-conditioning error, whereas text tokens carry far more information per position, so the fine grid is doing real work. On that account the divergence should move with tokenizer bitrate; CC12M’s higher image optimum ($f8$, 1,024 positions; 16–20 versus 10 steps) is one weak data point in the predicted direction.

Other inference choices. Three further choices were measured on CC3M at 256 steps (hence far from the image optimum; these are lower bounds on image quality). *Churn* is strongly harmful for text→image—at guidance 3.0 and $n=5000$, FID 28.10 / CMMD 22.57 deterministic against

CC3M, text→image, gs 3.0						CC12M, text→image, gs 2.0				
Steps	NFE	FID	CMMD	SWD	CLIP	Steps	NFE	FID	CMMD	CLIP
4	8	28.55	22.71	0.729	27.45	8	16	24.97	20.90	26.29
6	12	24.79	19.13	0.662	27.86	12	24	22.21	18.20	26.51
8	16	24.28	18.81	0.653	27.99	16	32	21.68	17.84	26.56
10	20	24.00	18.76	0.647	28.04	20	40	21.96	17.79	26.54
12	24	24.03	18.95	0.647	28.05	24	48	22.22	17.84	26.57
16	32	24.68	19.40	0.647	28.05	32	64	22.81	18.21	26.60
32	64	26.13	20.85	0.661	28.03	48	96	23.58	18.90	26.58
64	128	27.21	21.59	0.671	28.00	256	512	27.22	21.25	26.45
128	256	27.79	22.18	0.676	27.96					
256	512	28.10	22.57	0.682	28.00					

Table 9: Step-count sweeps, $n=5000$. Both minima are bracketed on both sides: a shallow bowl (8–14 steps for CC3M, 16–24 for CC12M) with a steep wall below and a long monotone climb above. The gain over 256 steps is larger for CC12M (-5.54 FID / -3.46 CMMD) than for CC3M (-4.10 / -3.81); the optimum moving up with model and sequence size is the direction expected if the mechanism is error accumulation across steps, but two points do not establish a trend. CLIP is flat to within 0.1 between 8 and 256 steps for CC3M and cannot see the effect. At the full protocol ($n=30,000$, guidance 3.0) CC3M moves from FID 22.79 / CMMD 22.16 at 256 steps to 18.73 / 18.56 at 10 steps, i.e. -4.06 FID at 1/25.6 the cost.

Steps	CIDEr↑	SPICE↑	BLEU-4↑	METEOR↑	ROUGE-L↑	i2t CLIP↑
10	0.0131	0.0088	0.0040	0.0483	0.1495	23.91
32	0.0588	0.0259	0.0139	0.0679	0.1807	24.34
64	0.0488	0.0233	0.0128	0.0665	0.1769	24.53
128	0.1304	0.0534	0.0285	0.0965	0.2192	—
256	0.1362	0.0541	0.0291	0.0975	0.2212	24.83
512	0.1032	0.0435	0.0237	0.0861	0.2019	24.90

Table 10: CC3M, text-emitting directions versus step count. Captioning is the complete Karpathy test split ($n=5000$, guidance 2.0, churn $\gamma=0.175$); i2t CLIP is the COCO reference set ($n=5000$, guidance 3.0). Captioning peaks at 256 steps and is bracketed on both sides, retaining 10% of its CIDEr at 10 steps; the 64-step point falling below 32 puts the run-to-run noise at roughly 0.01 CIDEr. Image→text CLIP rises monotonically and saturates (the last doubling buys 0.07). In the joint direction both halves come from one trajectory and move in opposite directions with the step count (CC3M, $n=5000$, 10 / 32 / 64 steps: FID 56.06 / 59.14 / 62.66, pair-CLIP 23.16 / 24.71 / 25.17); CC12M shows the same asymmetry (joint CMMD 37.60 at 20 steps versus 40.33 at 256, pair-CLIP 23.19 versus 24.01).

36.89 / 34.73 at $\gamma=0.1$ and 41.87 / 40.51 at $\gamma=0.175$, while CLIP moves by at most 0.06—yet the same $\gamma=0.175$ is what makes captioning work (deterministic decoding is degenerate, CIDEr 0.018; Table 14). *Schedule*: for text→image the entropy-derived and Karras grids agree at the guidance values actually used, 0.03–0.14 FID apart at guidance 3–5 (28.10 versus 27.96 at guidance 3.0), and separate below them, reaching 0.82 FID in Karras’s favour at guidance 1.0; the two arms ran two days apart on code that logged no sampler configuration, so we do not read that gap either way. For captioning the entropy-derived grid is decisively better, CIDEr 0.136 versus 0.049 and SPICE 0.054 versus 0.023 at guidance 2.0, so a text→image-only ablation would have concluded the schedule does not matter. *Self-conditioning*: carry matches refined at guidance 3.0 (CMMD 22.30 versus 22.44, CLIP 27.98 versus 27.99, both at $n=2000$) at half the NFE, so carry is used throughout. The flat CLIP columns of Tables 9 and 11, the churn result and the step-invariance of GenEval (§A.5) are four independent demonstrations that these metrics cannot arbitrate image quality for this model.

Guidance	CC3M			CC12M		
	FID	CMMD	CLIP	FID	CMMD	CLIP
1.0	33.65	25.80	25.32	31.29	22.32	24.80
1.5	29.71	23.10	26.51	27.80	21.39	25.82
2.0	28.61	22.44	27.19	27.22	21.25	26.45
3.0	28.10	22.57	28.00	27.46	22.32	27.10
4.0	28.06	22.71	28.44	28.53	23.24	27.42
5.0	28.19	23.27	28.72	—	—	—

Table 11: Guidance sweeps, text→image, 256 steps, $n=5000$ (noise paired across guidance values for CC3M). CMMD minimises near guidance 2, FID one to two notches higher, and CLIP rises monotonically throughout, so past the peak image quality must be judged on CMMD rather than CLIP. Both optima are bracketed on both sides. Guidance and step count are close to separable: re-sweeping at the re-tuned step counts leaves the structure unchanged (CC3M at 10 steps, guidance 2.0 / 2.5 / 3.0: CMMD 18.77 / **18.50** / 18.76; CC12M at 20 steps, guidance 1.5 / 2.0 / 2.5 / 3.0: CMMD 18.70 / **17.79** / 17.92 / 18.33), which is why the one-dimensional step sweep at fixed guidance was a valid search. CC12M loses CLIP to CC3M at every matched guidance value (by 0.52–1.02) while winning FID and CMMD at guidance 1–3.

Text→image, $n=30,000$	CC3M	CC12M	Δ
operating point	10 steps, gs 2.5	16 steps, gs 2.0	
FID (raw) ↓	18.48	16.59	−1.89
sampling floor (disjoint real draws)	2.35	2.35	0
codec contribution (unpaired)	0.59	0.20	−0.39
FID net of the codec	17.90	16.39	−1.50
CMMD ↓	18.41	17.76	−0.65
SWD ↓	0.632	0.591	−0.041
CLIP B/32 ↑	27.73	26.51	−1.22 (<i>worse</i>)
<i>Text-emitting directions, matched 256 steps, $n=5000$</i>			
captioning CIDEr ↑	0.1362	0.0571	2.4× <i>worse</i>
captioning SPICE ↑	0.0541	0.0347	1.6× <i>worse</i>
captioning BLEU-4 ↑	0.0291	0.0120	2.4× <i>worse</i>
captioning METEOR ↑	0.0975	0.0752	1.3× <i>worse</i>
i2t CLIP B/32, gs 3.0 ↑	24.83	22.98	−1.85
i2t CLIP B/32, gs 4.0 ↑	24.97	23.08	−1.89

Table 12: Scaling verdict at each model’s own optimum. On the image side CC12M wins every distributional metric and the margin survives floor correction: removing each tokenizer’s codec contribution, measured against a disjoint real draw as the generative FID is (§A.1), leaves CC12M 1.50 ahead, so the codec explains about a fifth of the raw gap rather than half (Fréchet distances are not additive, so this remains a reporting heuristic showing the order of the confound); the shared 2.35 sampling floor cancels. No reconstruction-CMMD floor has been measured, so the CMMD margin retains an unquantified codec advantage. On the text side CC3M wins every metric by margins far outside the ~ 0.01 CIDEr run-to-run noise. The image-side gap widens as the sampler approaches its optimum: at a matched 32 steps and guidance 2.0 ($n=5000$) it is 3.27 FID (26.09 versus 22.81) against 1.39 at 256 steps, so the earlier 256-step comparison was masking CC12M’s advantage.

A.4 CC3M VERSUS CC12M

Interpretation, and what it rests on. Scaling the corpus 4×, with the other advantages of Table 7, buys 1.89 FID on text→image (1.50 after floor correction) and costs about 60% of captioning CIDEr. Undertraining (CC12M stopped at 87% of its budget with a 3.6× longer sequence) predicts a uniform deficit; noisier image–text correspondence in CC12M’s alt-text predicts the split observed, since more images improve the marginal image distribution while weaker caption–image alignment degrades everything conditional on text. The measured pattern is the second. This is an inference from a pattern across metrics, not a direct measurement of caption quality: the decisive test, a CMMD-versus-checkpoint curve over CC12M’s saved checkpoints, has not been run.

Model	Params	Single	Two	Count	Colors	Pos.	Attr.	Overall
Show-o (Xie et al., 2024)	1.3B	0.98	0.80	0.66	0.84	0.31	0.50	0.68
SD3	2B	0.98	0.74	0.63	0.67	0.34	0.36	0.62
SD-XL	—	0.98	0.74	0.39	0.85	0.15	0.23	0.55
SD v2.1	—	0.98	0.51	0.44	0.85	0.07	0.17	0.50
SD v1.5	—	0.97	0.38	0.35	0.76	0.04	0.06	0.43
CLIP retrieval	—	0.89	0.22	0.37	0.62	0.03	0.00	0.35
LlamaGen	0.8B	0.71	0.34	0.21	0.58	0.07	0.04	0.32
CoBit-CC3M (ours)	462M	0.80	0.12	0.05	0.46	0.04	0.03	0.249
minDALL-E	—	0.73	0.11	0.12	0.37	0.02	0.01	0.23
CoBit-CC12M (ours)	633M	0.68	0.11	0.06	0.45	0.01	0.01	0.222

Table 13: Per-task GenEval scores; prior work as printed in Ghosh et al. (2023) (Table 2) and Xie et al. (2024) (Table 3). Our rows use the entropy-derived grid, deterministic sampling, guidance 3.0 and 256 steps. The score is flat in the step count—at each model’s image optimum CC3M scores 0.235 (10 steps) and CC12M 0.221 (16)—so GenEval does not register the 4.06 FID between those settings. Single objects render 80% of the time and colours 46%, while compositional binding stays near the floor (attribute 0.03, position 0.04), as it does for every sub-billion-parameter model in the table; CC3M alt-text rarely specifies spatial relations or colour-object bindings. The sampler configuration matters materially: an earlier pass of the same checkpoint that had silently fallen back to a Karras grid at guidance 2.0 scored 0.209.

Configuration	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	CLIPScore	RefCLIPScore	CLIP L/14
<i>CC3M, 256 steps</i>								
deterministic, gs 1.0 (<i>degenerate</i>)	0.004	0.054	0.157	0.018	0.012	—	—	—
entropic, churn $\gamma=0.175$, gs 4.0	0.0147	0.0695	0.1804	0.0520	0.0247	0.623	0.6678	18.34
entropic, churn $\gamma=0.175$, gs 2.0	0.0291	0.0975	0.2212	0.1362	0.0541	0.612	0.6606	17.86
Karras, churn $\gamma=0.175$, gs 2.0	0.0131	0.0672	0.1792	0.0485	0.0232	—	0.6578	—
<i>CC12M, 256 steps</i>								
entropic, churn $\gamma=0.175$, gs 2.0	0.0120	0.0752	0.1719	0.0571	0.0347	—	—	—

Table 14: Zero-shot captioning on the complete Karpathy test split ($n=5000$, no empty captions). Guidance 2.0 maximises the n -gram metrics and guidance 4.0 the CLIP-space metrics; the two families disagree, so both are reported. The entropy-derived grid beats Karras by $2.81\times$ on CIDEr while RefCLIPScore moves by 0.003.

A.5 GENEVAL BREAKDOWN

A.6 CAPTIONING

No published zero-shot baseline exists for this number. UniDiffuser (Bao et al., 2023a) evaluates image→text with CLIP score only, and L-Verse’s captioning numbers (Kim et al., 2022) are for a model trained on the MS-COCO Karpathy splits. For scale, on the $\times 100$ convention those papers use, L-Verse reaches BLEU-4 39.9 / METEOR 31.4 / ROUGE 60.4 / CIDEr 102.2 / SPICE 23.3, M^2 Transformer (Cornia et al., 2020) CIDEr 131.2 and OSCAR_L (Li et al., 2020) 140.0, all trained on MS-COCO; our 0.136 is roughly $7.5\times$ below L-Verse, not $100\times$. L-Verse’s own no-truncation variant moves CIDEr from 102.2 to 181.6 while BLEU-4 falls from 39.9 to 27.6 on the same model and images, so caption length and phrasing alone swing these metrics by tens of points: they track surface form at least as much as content. That is the failure here. The model writes fluent CC3M-style alt-text (“*example of a classic bathroom design with stainless countertops*”) against COCO references written as “*A bathroom with a shower and a sink.*”; CIDEr’s IDF-weighted n -grams penalise the register mismatch almost regardless of content. We report CIDEr for completeness and read the result through RefCLIPScore.

A.7 NOTES ON COMPARABILITY AND MEASUREMENT HYGIENE

Literature figures. Every number in Table 3 was read from the cited paper’s own table or from the compilation tables of Xie et al. (2024), Bao et al. (2023a) and Zhou et al. (2022). Improved VQ-Diffusion (Tang et al., 2022), our closest architectural relative (discrete diffusion over VQ tokens,

The task is evaluating the relevance and likelihood of the predicted text continuation, given the text prompt. You should also consider whether the meaning of the text continuation is making sense. The text prompt is: "prompt", and the text continuation is: "content"

You must give an overall rating from 1 to 5. The rating guideline is as below:

- 1: The text continuation is very unlikely and irrelevant to the text prompt.
- 2: The text continuation is unlikely and marginally relevant to the text prompt.
- 3: The text continuation is moderately likely and relevant to the text prompt.
- 4: The text continuation is likely and relevant to the text.
- 5: The text continuation is very likely and highly relevant.

You should take the following steps to provide the score:

First: briefly analyse the sample with the above definition.

Second: MUST follow the output format as: I would rate the score as _

Table 15: LLM-as-a-Judge prompt for thematic relevance evaluation of speech continuation. GPT-4o is used as the judge model.

trained on CC3M+CC12M), reports FID 8.44 on MS-COCO and 15.58 on Conceptual Captions; neither is a zero-shot COCO number (its COCO table compares against COCO-trained models, and the CC figure is in-domain), so it is excluded. DALL·E’s 27.50 is a best-of-512 CLIP-reranked figure its own paper does not print, reproduced here as the compilation tables give it; CogView’s 27.10 is excluded because that evaluation blurs MS-COCO, translates the captions and re-ranks 60 samples each. Only the ViT-L/14 column of Table 8 may be placed beside published CLIP scores; the B/32 and L/14 scales differ substantially (28.02 versus 21.79 for the same samples). Generations faithfully reproduce CC3M’s distribution (stock-photo watermarks, white-background product shots, single subjects) against COCO’s clean multi-object scenes; that distance penalises FID independently of image quality, and Xie et al. (2024) make the same observation from the other side for models fine-tuned on aesthetic data.

Schedule audit. If the entropy tables are absent from a run directory the sampler silently reverts to a Karras grid. Every run reported here was audited for the corresponding warning; every number reported here comes from a run with a count of zero. Where the fallback did occur, the run either supplies a “Karras” arm of the schedule comparison, in which the fallback is the treatment, or was superseded by a fallback-free repeat whose numbers are the ones printed. The audit is not optional: the unaudited GenEval pass that scored 0.209 had fallen back.

B AUDIO AND TEXT EXPERIMENTS

B.1 PROTOCOL AND CHECKPOINTS

LibriSpeech-PC. To assess the linguistic properties of generated speech signals, we transcribe them using an external Whisper-large (Radford et al., 2023) ASR model. For co-generation, error rates are computed between the two generated sequences. The fluency of a text sequence is measured by its perplexity under an external language model (in this case Llama-3.2 1B), computed separately for each modality in co-generation and for speech transcriptions in speech continuation. To further determine the conceptual relevance of produced continuations, GPT-4o is employed in an LLM-as-a-Judge setting, assigning each continuation a score from 1 to 5. The system message is taken directly from Tseng et al. (2025) and presented in Table 15. For TTS, voice replication is captured in terms of speaker similarity (SpkSim), computed as the cosine similarity between the outputs of an ECAPA-TDNN (Desplanques et al., 2020) speaker verification model.

Distributional Metrics. For speech continuation, we are also interested in how closely the generated audio follows the distribution of ground-truth continuations. We thus measure Fréchet Speech Distance (FSD), an adaptation of the FID metric to audio. Population statistics are extracted from self-supervised representations of the true and generated continuations, and their Fréchet distance is calculated. Following previous work (Chou et al., 2025), two variants are considered: FSD-wlm, which is based on embeddings from the 6th layer of WavLM-large (Chen et al., 2022) to measure frame-level match, and FSD-e2v, which is based on expressiveness representations from

Direction	Steps (NFE)	γ	gs
<i>CoBit-LibriTTS — one set of weights (134M, 800 speech tokens)</i>			
text $\rightarrow$ speech	512 (1024)	0.23	6
speech $\rightarrow$ text	512 (1024)	0.37	5
speech continuation	256 (512)	0.17	5
joint	256 (256)	0.11	—
<i>CoBit-MLS — one set of weights (633M, 500 speech tokens)</i>			
text $\rightarrow$ speech	512 (1024)	0.27	5
speech $\rightarrow$ text	512 (1024)	0.27	5
speech continuation	256 (512)	0.28	6
joint	512 (512)	0.23	—

Table 16: Sampler operating points for text-audio experiments. Steps, stochasticity and guidance settings were chosen based on validation performance on 500 samples from LibriTTS dev-clean.

Emotion2Vec (Ma et al., 2024). FSD metrics do not evaluate the quality of the generated audio; they capture its population-level resemblance to the original continuations.

Generation-based SALMON for acoustic evaluation. The SALMON benchmark is used for acoustic evaluation. In its original form, SALMON creates replicas of short speech recordings where the start remains identical, but some paralinguistic property (sentiment, speaker identity/gender, background, room) is modified midway. SLMs receive an accuracy score related to how often the positive example is assigned higher likelihood than the negative one. Given the incompatibility of likelihood-based evaluation for diffusion models (Theis et al., 2015), we turn to an alternative generation-based SALMON formulation (Hsu et al., 2026). The examined SLM produces a continuation of the shared start of the two examples, and a judge model calculates embeddings for the positive, negative and generated continuations. The SLM then receives an accuracy score based on how often the generated continuation is closer to the positive example than the negative one, in terms of cosine similarity in the embedding space. TITANET (Koluguri et al., 2022) is used as the judge for sentiment, speaker and gender tasks; HuBERT-large-audioset and Wav2Vec2-large-audioset (La Quatra et al., 2024) are used for background and room tasks, respectively.

Operating points. The sampler configurations (integration steps, stochasticity and guidance scale) for each conditional task have been selected based on modality alignment (WER) on 500 samples from the LibriTTS dev-clean set; the final settings are reported in Table 16. Samplers with 256 and 512 neural function evaluations (NFE) are considered separately, with stochasticity $\gamma \in \{0.11, 0.13, 0.15, 0.17, 0.19, 0.22, 0.25, 0.28\}$ for $\text{NFE} = 256$ and $\gamma \in \{0.15, 0.19, 0.23, 0.27, 0.31, 0.34, 0.37, 0.4\}$ for $\text{NFE} = 512$. In all cases, guidance scales $\text{gs} = 5$ and $\text{gs} = 6$ were examined. Higher stochasticity values were preferred for $\text{NFE} = 512$ in comparison to $\text{NFE} = 256$.

B.2 GROUND-TRUTH SPEAKER FEATURES AND TRAINING-INFERENC MISMATCH

For all three conditional tasks, BiCodec-global speaker tokens are part of the condition. For ASR, speaker tokens are extracted from the full speech signal, which is always provided as input. Inference-time speaker tokens are extracted from the audio prompt for continuation and from a held-out recording for TTS to rule out information leakage. This is standard practice for zero-shot TTS. During training, however, speaker features from the full ground-truth audio samples are used for all tasks. This forces CoBit to learn a tighter relationship between speaker features and speech tokens, simultaneously accounting for the variability in BiCodec-global, which does not guarantee that all recordings of the same speaker produce identical outputs. A training-inference mismatch is therefore introduced to facilitate unified training.

B.3 COMPLETE RESULTS

Tables 17 and 18 contain extensive evaluation metrics and baseline comparisons for co-generation and cross-modal directions, respectively; Table 19 details per-task SALMON accuracy. LLM-as-a-Judge score distributions for CoBit-LibriTTS and CoBit-MLS are visualized in Figure 4

Model	Params	WER↓	CER↓	UTMOS↑	GenPPL-Text↓	GenPPL-Speech↓
<i>CoBit (ours; from scratch, no pretrained components)</i>						
CoBit-LibriTTS	134M	20.2	13.2	3.02	668.1	395.6
CoBit-MLS	633M	3.7	2.3	3.78	280	141.2
<i>Autoregressive System (SmoLLM+Speaker → SparkTTS)</i>						
Cascade	640M	3.2	3.1	4.23	122	1,648

Table 17: Joint generation performance, computed by generating 1,024 samples. Speech perplexity for the cascade system is high, as SmoLLM produces multilingual sequences or web/code artifacts that are not transferable to speech. We use the same NFE as in Table 4; 256 for CoBit-LibriTTS and 512 for CoBit-MLS.

Model	Params	Text-to-Speech				Speech-to-Text	
		WER↓	CER↓	UTMOS↑	SpkSim↑	WER↓ Clean	WER↓ Other
<i>CoBit (ours; from scratch, no pretrained components)</i>							
CoBit-LibriTTS	134M	11.7	8.1	3.31	0.19	23.3	38
CoBit-MLS	633M	5.3	4.1	4	0.27	8.2	22
<i>Autoregressive SLMs/Hybrids</i>							
GPA-v1.5 (Cai et al., 2026)	505M	3.1	3.1	4.37	0.4	3.4	6.3
SpiritLM-Base	7B	—	6.7	—	—	6	11
Moshi (Défossez et al., 2024)	7B	4.7	—	—	—	5.7	—

Table 18: Text-to-speech and Speech-to-text performance on LibriSpeech-PC. For text-to-speech, evaluation is conducted on the test-clean partition. For each speaker in the dataset, one sample was held out to extract speaker features. Speech-to-text evaluation is conducted on test-clean and test-other partitions separately. As in Table 4, NFE=512 for both CoBit-LibriTTS and CoBit-MLS.

B.4 SAMPLER CHARACTERISATION

[Decide what settings need to be considered to match image/biological experiments]

B.5 BASELINE SYSTEMS

All known speech language models (SLMs) are autoregressive; diffusion models in speech generation are mostly limited to TTS systems or ASR-TTS hybrids.

Multimodal SLMs and Hybrids. The most comparable text-audio baselines for CoBit are models that learn to generate both text and speech. TASLM (Tseng et al., 2025) generates the two modalities jointly, leveraging continuous text-aligned speech representations from the TASTE encoder and implementing separate prediction heads. The semantic equivalence of text units and speech embeddings bridges the gap between audio and text generation, producing linguistically fluent and conceptually consistent audio. Other approaches focus on systems that can be extended to cross-modal tasks. SpiritLM (Nguyen et al., 2025) is trained on interleaved text-audio sequences, and its expressive variant for additional control over pitch and style. SpiritLM models can perform TTS and ASR via task-specific fine-tuning and few-shot prompting. Moshi (Défossez et al., 2024) is a spoken dialogue system that produces text alongside speech, as an inner monologue that steers generation towards intelligible sentences. Its cross-modal capabilities arise from introducing a time delay on the generated stream and inference-time teacher-forcing on the conditioning stream. GPA (Cai et al., 2026) is a multi-task autoregressive hybrid that treats TTS and ASR as conditional language modelling problems. It is the closest method to CoBit in terms of scale, multi-task joint training and initialization from scratch, but it is limited to cross-conditional tasks and cannot perform speech language modelling. However, all of these models require task-specific design and modifications for TTS and ASR, in contrast to CoBit.

Speech-only SLMs. In the context of speech generation, additional comparisons can be drawn to textless SLMs, which mainly rely on the powerful Mimi tokenizer to computationally balance

Model	Sentiment	Speaker	Gender	Bg.- Domain	Bg.- Random	Room
<i>CoBit (ours; from scratch, no pretrained components)</i>						
CoBit-LibriTTS	89	88.5	91	65.5	74	73.5
CoBit-MLS	90	94.5	97	69	81	74
<i>Trained on MLS-English (same corpus as CoBit-MLS, excluding LibriTTS)</i>						
Flow-SLM-S	85	87	92.5	77	86	69.5
Flow-SLM-L	92.5	99.5	99	77	84.5	77
<i>Speech-only SLMs</i>						
Flow-SLM-Ext.	93	98	100	77	86.5	85.5
Llama-Mimi	93	90	95	81	90	78.5
<i>Text+Speech SLMs</i>						
TASLM-Embed	95	100	100	61.5	54.5	57.5
SpiritLM-Base	47.5	52	50.5	50.5	52.5	50
SpiritLM-Expressive	72	54	56	58.5	55.5	54.5

Table 19: Speech continuation performance across the consistency tasks of the SALMON benchmark for generation-based evaluation. Each model generates a continuation for each prompt, and accuracy scores are computed for each task by comparing the cosine similarity of the generated continuation with the positive and negative examples. Different embedding models are used for each task, specified in Appendix B.5. NFE256 is used for both CoBit-LibriTTS and CoBit-MLS.

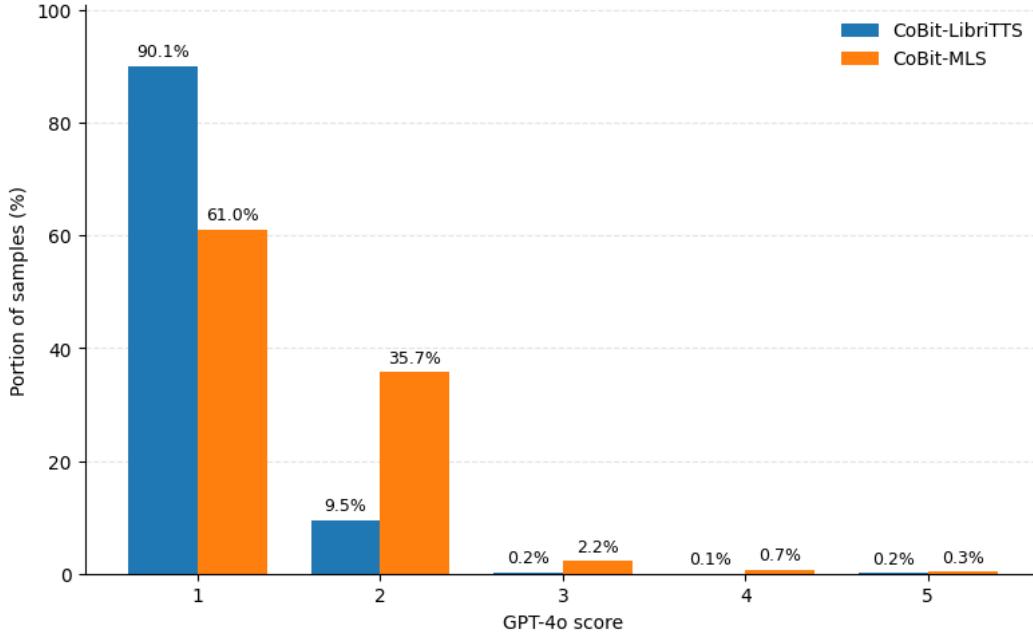


Figure 4: LLM-as-a-Judge score distribution for CoBit-LibriTTS and CoBit-MLS. CoBit-LibriTTS almost always receives the lowest possible score. A large portion of CoBit-MLS samples perform moderately better, but neither of the model can consistently achieve scores of 3 or higher.

the semantic benefits of text generation with fine-grained acoustic details. Llama-Mimi (Sugiura et al., 2025) autoregressively predicts each Mimi layer, leveraging RVQ’s conditional nature: the generation of each code is conditioned on both the previous tokens and the earlier codebooks of the same token. Flow-SLM (Chou et al., 2025) exploits Mimi’s semantic-acoustic combination for cascaded prediction, breaking the strictly autoregressive paradigm: a causal transformer predicts the semantically distilled first RVQ layer; then, unquantized Mimi embeddings are inferred with flow matching, conditional on look-ahead semantic tokens.

Cascaded Co-generation. None of the multimodal systems are evaluated on unconditional co-generation. TASLM-Embed and Moshi are the only baselines that are, in theory, capable of true parallel generation, but they are not trained for such a task. Matching the autoregressive nature of all other baselines, we instead compare CoBit to a cascaded pipeline: text is generated unconditionally with a text LM and speaker identity is sampled from a small transformer; a TTS system then uses the text and speaker features to synthesize the speech sequence. This cascade system effectively targets the same distribution as CoBit for joint generation, factorized as $p(\text{text}, \text{speaker}, \text{speech}) = p(\text{text})p(\text{speaker})p(\text{speech}|\text{text}, \text{speaker})$. To match the 630M parameter scale of CoBit, SmoLLM2-135M (Allal et al., 2025) is used for text generation and Spark-TTS (Wang et al., 2025a) is used for speech synthesis. Speaker features are sampled by a transformer with 1M parameters in the form of BiCodec-global tokens and passed to the Spark-TTS decoder. We train the speaker transformer on LibriTTS and MLS-English. All three components of the cascade are autoregressive. A known drawback is that unconditional generation with a generic text LM can lead to web or code artifacts that cannot be rendered into natural speech, causing very high GenPPL-Speech, as observed in Table 17.

B.6 MEASUREMENT COMPARABILITY

For speech continuation and co-generation, we evaluate all baselines under the same protocol as CoBit using publicly available checkpoints. For cross-modal transfer, we only evaluate GPA-v1.5. While public SpiritLM and Moshi models do exist, their TTS and ASR performance relies on fine-tuning and time-delaying tricks (B.5) that cannot be exactly replicated; we therefore report evaluation numbers from their respective papers. All generation-based SALMON scores are taken directly from Hsu et al. (2026). However, the exact protocol documented in the original work is replicated, using the same checkpoints for the judge models.

C PROTEIN SEQUENCE AND STRUCTURE EXPERIMENTS

[Bruno: additional results and experimental details for reproducibility.]